Clinical Genomics

Iris Li & Michael Schatz
Oct 21, 2026

JHU EN.601.449/EN.601.649: Applied Comparative Genomics



Outline

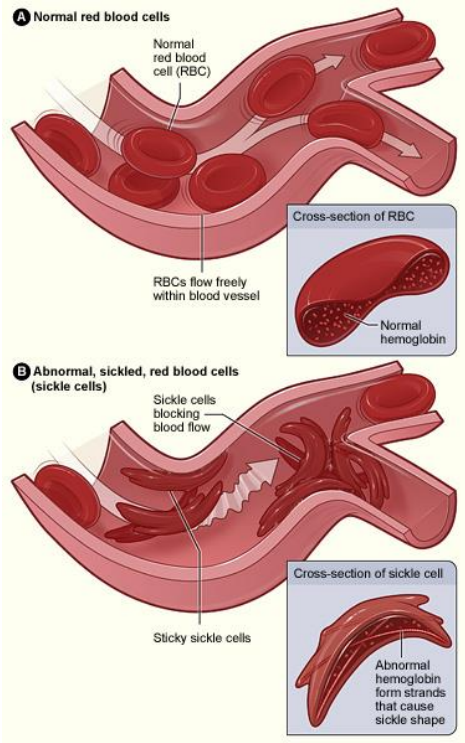
- Early discoveries in human disease genetics
- Genetic association studies and functional interpretation
- Population-scale long-read genomics in All of Us

Sickle Cell Anaemia

- **Sickle cell anemia** is an inherited blood disorder in which red blood cells become sickle-shaped and can obstruct blood flow.

- **A single A-to-T substitution in the HBB gene** changes glutamic acid to valine, producing hemoglobin S (HbS).

- **Autosomal recessive inheritance:** Individuals with two HbS alleles develop sickle cell anemia, while heterozygous carriers have some protection against severe malaria.



OMIM: SICKLE CELL ANEMIA
https://www.omim.org/entry/603903

Huntington's Disease

A polymorphic DNA marker genetically linked to Huntington's disease

James F. Gusella*, Nancy S. Wexler†‖, P. Michael Conneally†, Susan L. Naylor§, Mary Anne Anderson*, Rudolph E. Tanzi*, Paul C. Watkins*¶, Kathleen Ottina*, Margaret R. Wallace‡, Alan Y. Sakaguchi§, Anne B. Young‖, Ira Shoulson‖, Ernesto Bonilla‖ & Joseph B. Martin*

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‖ Venezuela Collaborative Huntington's Disease Project*

Family studies show that the Huntington's disease gene is linked to a polymorphic DNA marker that maps to human chromosome 4. The chromosomal localization of the Huntington's disease gene is the first step in using recombinant DNA technology to identify the primary genetic defect in this disorder.

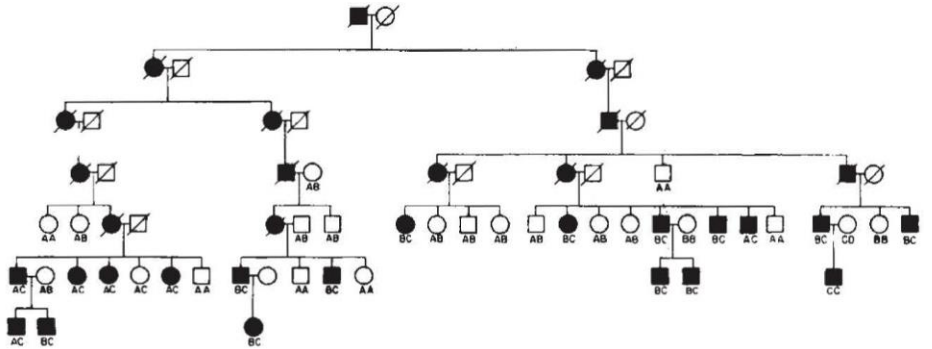
Gusella et al, Nature, 1983

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Family-based genetic evidence



Venezuelan Huntington's disease family

- Linkage analysis of a Venezuelan Huntington's disease family identified strong linkage to a DNA marker on chromosome 4.
- This localized the disease gene to chromosome 4, but the causal mutation remained unknown.

Gusella et al, Nature, 1983

Huntington's Disease

Cell, Vol. 72, 971–983, March 26, 1993, Copyright © 1993 by Cell Press

A Novel Gene Containing a Trinucleotide Repeat That Is Expanded and Unstable on Huntington's Disease Chromosomes

The Huntington's Disease Collaborative Research Group*

Summary

The Huntington's disease (HD) gene has been mapped in 4p16.3 but has eluded identification. We have used haplotype analysis of linkage disequilibrium to spotlight a small segment of 4p16.3 as the likely location of the defect. A new gene, IT15, isolated using cloned trapped exons from the target area contains a polymorphic trinucleotide repeat that is expanded and unstable on *HD* chromosomes. A $(CAG)_n$ repeat longer than the normal range was observed on *HD* chromosomes from all 75 disease families examined, comprising a variety of ethnic backgrounds and 4p16.3 haplotypes. The $(CAG)_n$ repeat appears to be located within the coding sequence of a predicted ~348 kd protein that is widely expressed but unrelated to any known gene. Thus, the *HD* mutation involves an unstable DNA segment, similar to those described in fragile X syndrome, spino-bulbar muscular atrophy, and myotonic dystrophy, acting in the context of a novel 4p16.3 gene to produce a dominant phenotype.

Introduction

Huntington's disease (HD) is a progressive neurodegenerative disorder characterized by motor disturbance, cognitive loss, and psychiatric manifestations (Martin and Gusella, 1986). It is inherited in an autosomal dominant fashion and affects ~1 in 10,000 individuals in most populations of European origin (Harper et al., 1991). The hallmark of HD is a distinctive choreic movement disorder that typically has a subtle, insidious onset in the fourth to fifth decade of life and gradually worsens over a course of 10 to 20 years until death. Occasionally, HD is expressed in juveniles, typically manifesting with more severe symptoms including rigidity and a more rapid course. Juvenile onset of HD is associated with a preponderance of paternal transmission of the disease allele. The neuropathology of HD also displays a distinctive pattern, with selective loss of neurons that is most severe in the caudate and putamen. The biochemical basis for neuronal death in HD has not yet been explained, and there is consequently no treatment effective in delaying or preventing the onset and progression of this devastating disorder.

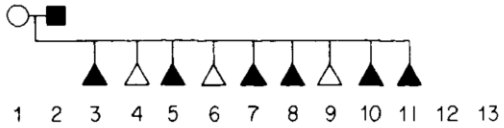
The genetic defect causing HD was assigned to chromosome 4 in 1983 in one of the first successful linkage analyses using polymorphic DNA markers in humans (Gusella

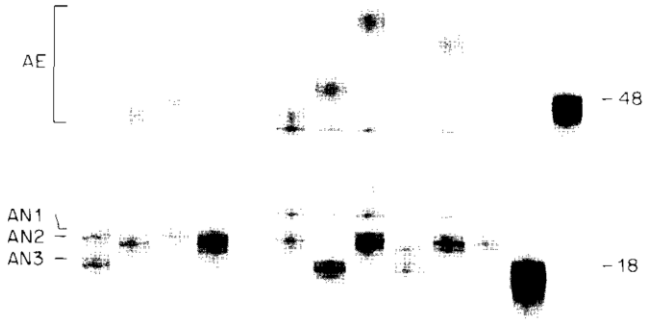
Huntington's Disease

Cell, Vol. 72, 971–983, March 26, 1993, Copyright © 1993 by Cell Press

A Novel Gene Containing a Trinucleotide Repeat That Is Expanded and Unstable on Huntington's Disease Chromosomes

CAG repeat expansion in a Venezuelan Huntington's disease family





Expanded CAG repeats in the HTT gene were found in all 75 Huntington's disease families studied, identifying the disease-causing mutation.

- AN1-AN3: Normal CAG repeat alleles of different lengths.
- AE: Expanded CAG repeat allele associated with Huntington's disease.

From Mendelian disease genes to complex traits

Human disease genes

Gerardo Jimenez-Sanchez*, Barton Childs* & David Valle*†

** Department of Pediatrics, McKusick-Nathans Institute of Genetic Medicine, and † Howard Hughes Medical Institute, Johns Hopkins University School of Medicine, Baltimore, Maryland 21205, USA*

The complete human genome sequence will facilitate the identification of all genes that contribute to disease. We propose that the functional classification of disease genes and their products will reveal general principles of human disease. We have determined functional categories for nearly 1,000 documented disease genes, and found striking correlations between the function of the gene product and features of disease, such as age of onset and mode of inheritance. As knowledge of disease genes grows, including those contributing to complex traits, more sophisticated analyses will be possible; their results will yield a deeper understanding of disease and an enhanced integration of medicine with biology.

To test the proposal that classifying disease genes and their products according to function will provide general insight into disease processes[1,2], we have compiled and classified a list of disease genes. To assemble the list, we began with 269 genes identified in a survey of the 7th edition of *Metabolic and Molecular Bases of Inherited Disease*[2]. We then searched the 'morbid map' and allelic variants listed in the Online *Mendelian Inheritance in Man*[3] (OMIM), an online resource documenting human diseases and their associated genes (www.ncbi.nlm.nih.gov), and increased the total disease gene set to 923. This sample included genes that cause monogenic disease (97% of the sample) and genes that increase susceptibility for complex traits. We excluded genes associated only with somatic genetic disease (such as non-inherited forms of cancer) or the mitochondrial genome.

Functional classification

We categorized each disease gene according to the function of its

- In 2001, researchers catalogued 923 human disease genes, 97% of which were associated with monogenic disorders.
- The genetic contributors to complex traits were still much less understood.

Jimenez-Sanchez, G., Childs, B. & Valle, D. Nature, 2001



Outline

- Early discoveries in human disease genetics
- Genetic association studies and functional interpretation
- Population-scale long-read genomics in All of Us

Genetic association studies and functional interpretation

1. How can we systematically identify variants associated with complex traits?
Genome-wide association studies (GWAS)

2. Among many associated variants, which are most likely causal?
Fine-mapping

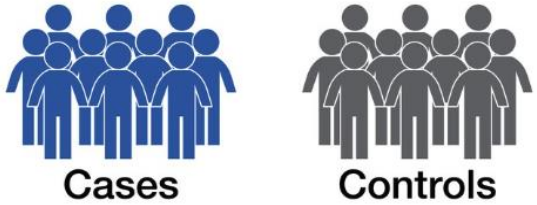
3. Beyond trait associations, how does genetic variation relate to gene expression?
Expression quantitative trait loci (eQTL)

Genome-wide association study (GWAS)

A genome-wide search for genetic variants associated with a trait

1 Sample selection

Trait example: Type 2 diabetes



Who has the trait?

2 Scan the genome

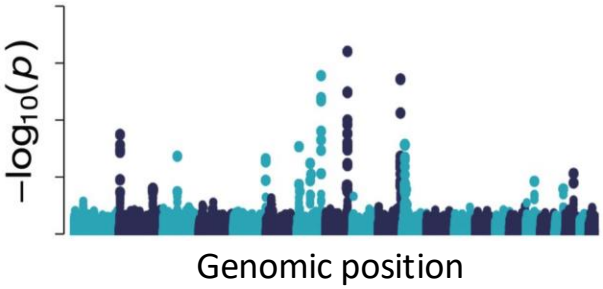
Genotypes at many variants



Genotype variants across the genome

3 Association testing

Manhattan plot



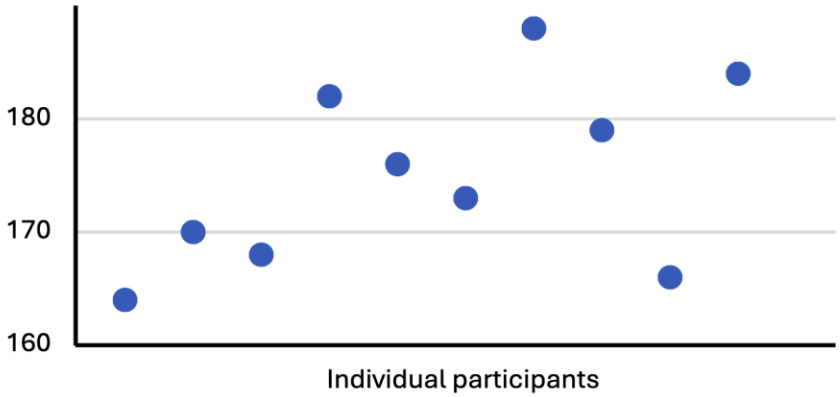
Which variants are associated with the trait?

GWAS association models depend on the phenotype type

Quantitative traits

Measured on a continuous numerical scale

Example: Height (cm)

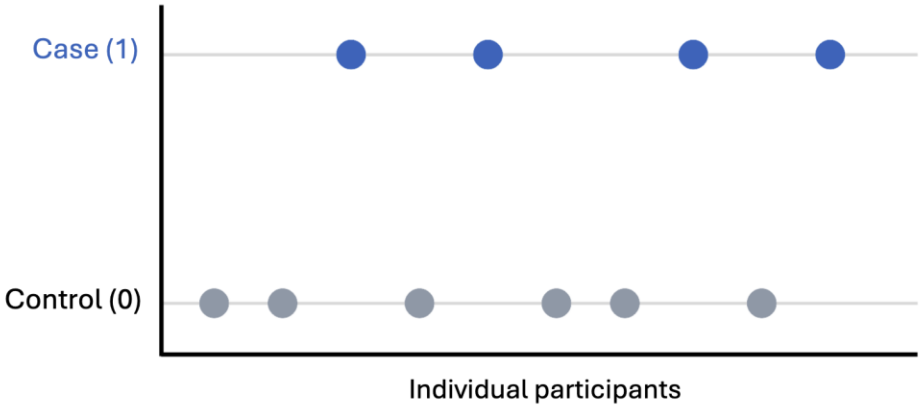


Association model: Linear regression

Binary traits

Two status: case or control

Example: Type 2 diabetes status

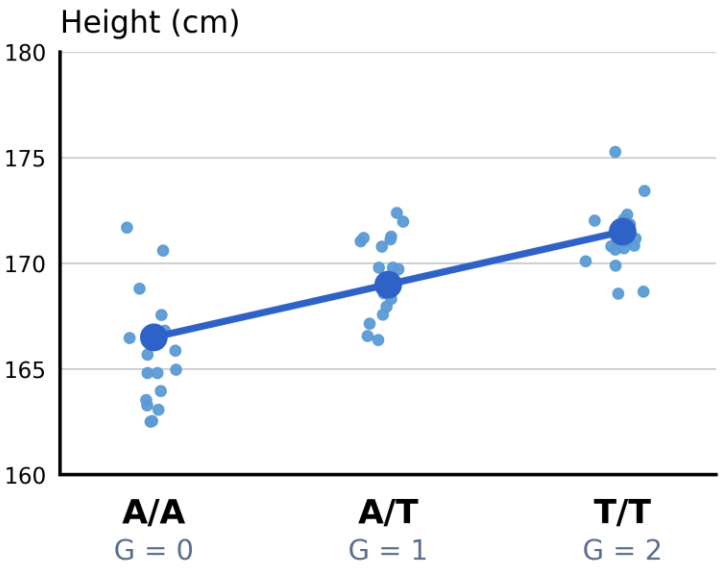


Association model: Logistic regression

Linear regression for quantitative traits

Is effect-allele dosage associated with a quantitative phenotype (e.g., height, blood pressure)?

Example: association between variant and height



Effect allele: T; Each dot: one participant

Additive association model

$$Y_i = \alpha + \beta G_i + \gamma^T X_i + \varepsilon_i$$

i denotes an individual

Y_i Quantitative phenotype

G_i Number of effect alleles (0, 1, or 2)
A/A = 0, A/T = 1, T/T = 2

β Mean trait difference per additional allele, adjusted for covariates

X_i Covariates (e.g., Age, sex, ancestry principal components)

α: intercept γ: covariate effects ε: residual

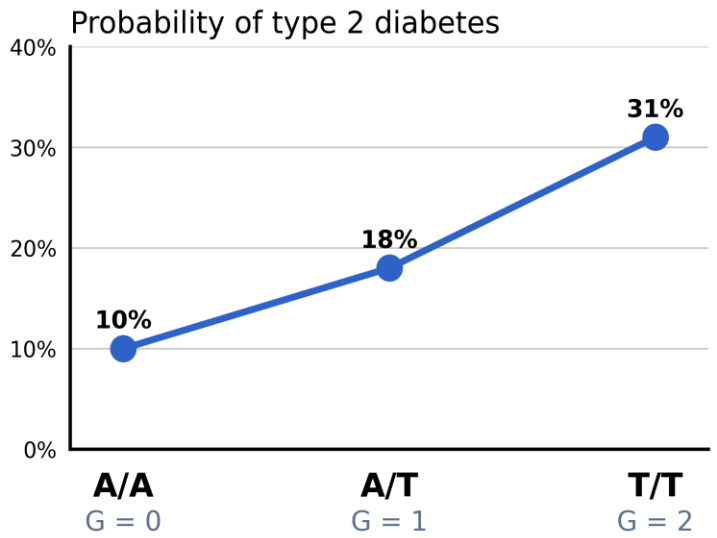
Association test $H_0: \beta = 0$

Report β, standard error (β), p-value

Logistic regression for binary traits

Is effect-allele dosage associated with a binary phenotype (e.g., type 2 diabetes)?

Example: association between variant and type 2 diabetes



Simulated: baseline probability = 10%, per-allele OR = 2

Additive logistic regression model

$$\text{logit}(p_i) = \alpha + \beta G_i + \gamma^T X_i$$

i denotes an individual

p_i	Probability of disease
G_i	Effect-allele dosage (0, 1, or 2)
β	Log-odds ratio per additional allele
X_i	Covariates (e.g., Age, sex, ancestry principal components)

α: intercept γ: covariate effects

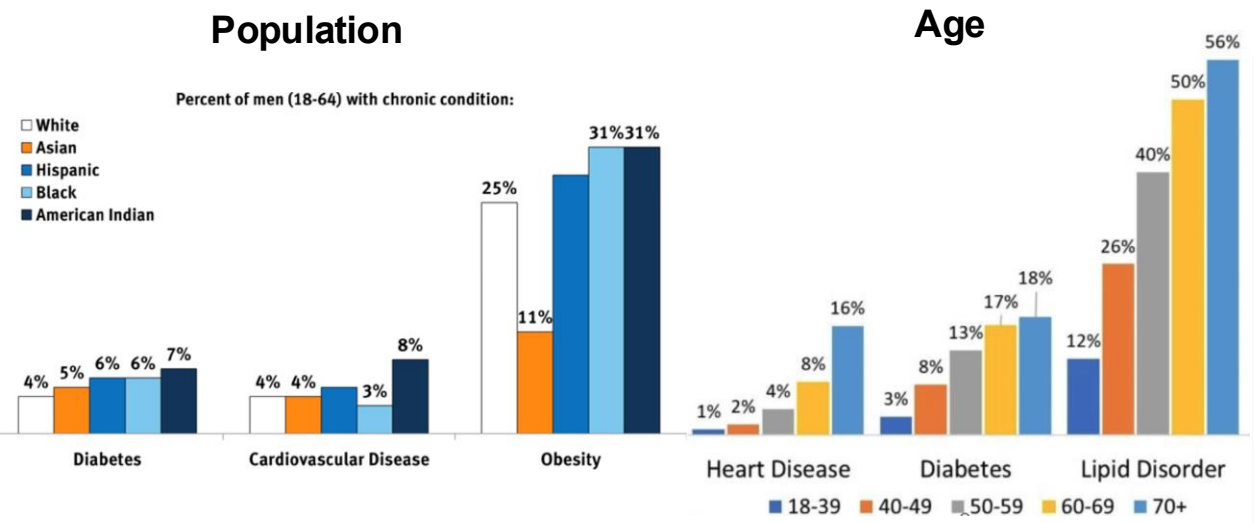
Per-allele odds ratio: odds ratio (OR) = e^{β}

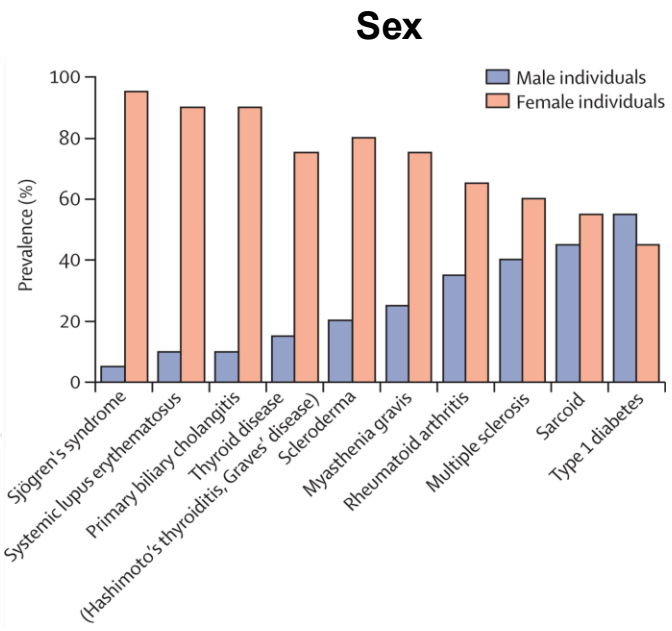
Test no association: H_0: β = 0

Report OR, 95% confidence interval (CI), p-value

Covariates and confounding in GWAS

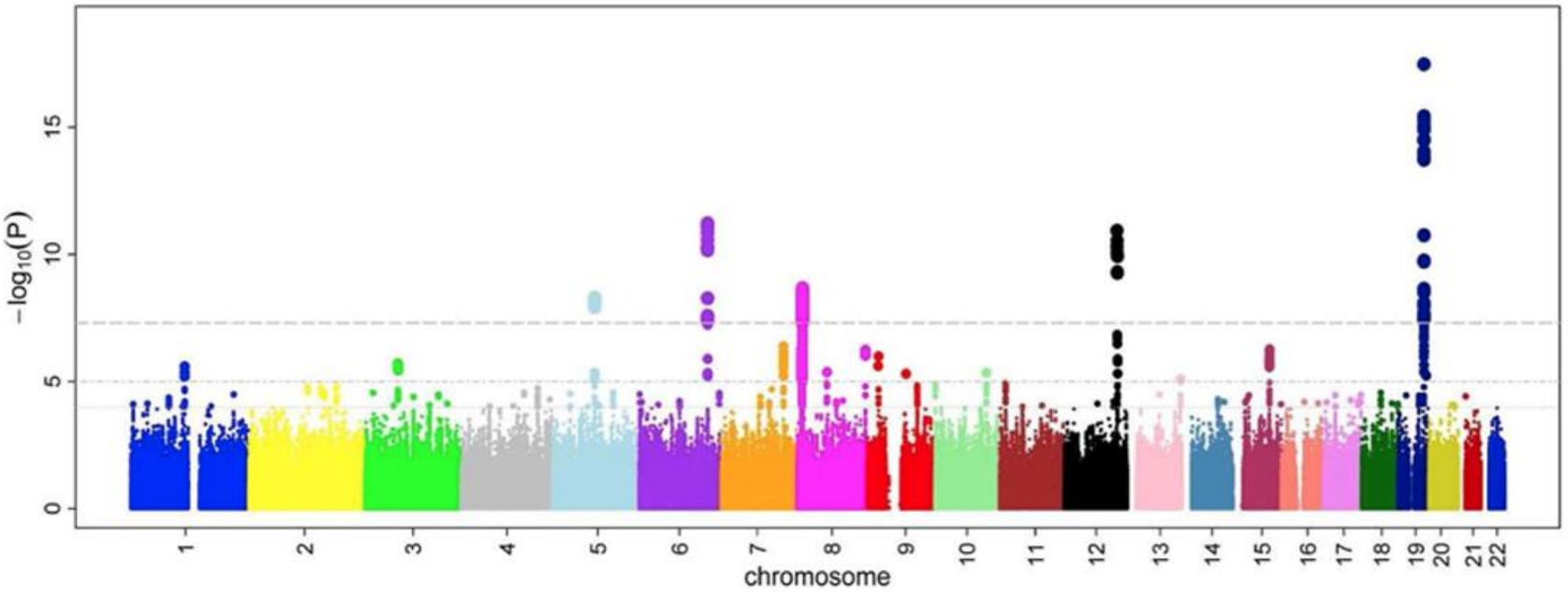
- Disease patterns differ across demographic groups





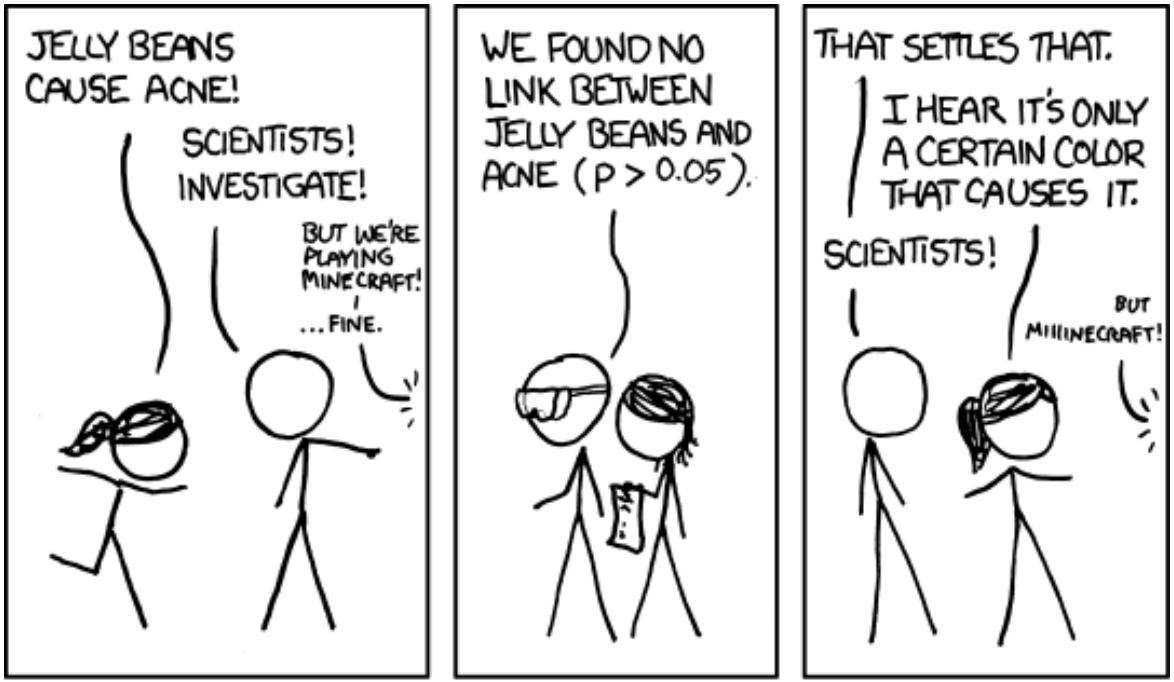
- Disease prevalence varies by age, sex, and population group.
- Adjusting for relevant covariates helps reduce confounding and the risk of false-positive associations in GWAS.

Manhattan Plot

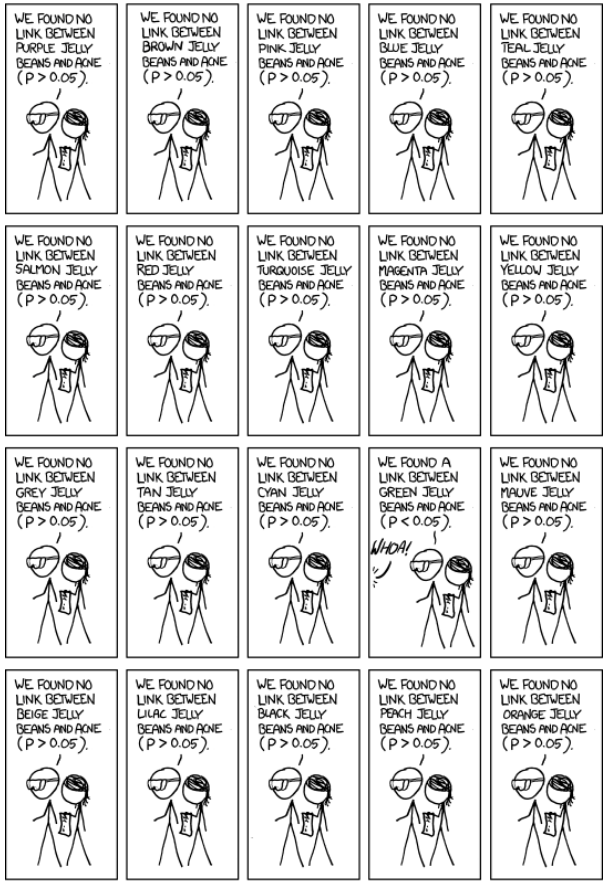


Four Novel Loci (19q13, 6q24, 12q24, and 5q14) Influence the Microcirculation In Vivo
Ikram et al (2010) PLOS Genetics. doi: 10.1371/journal.pgen.1001184

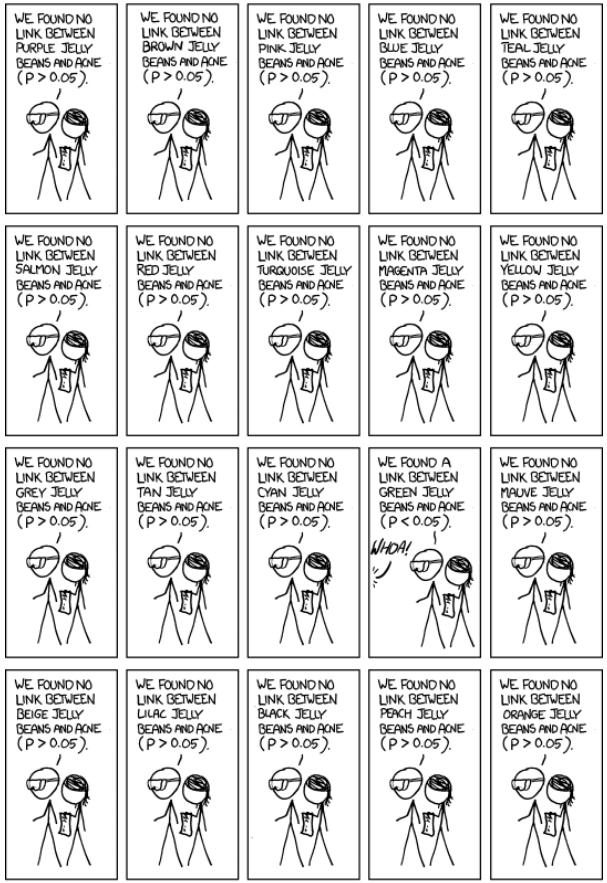
The curse of multiple testing

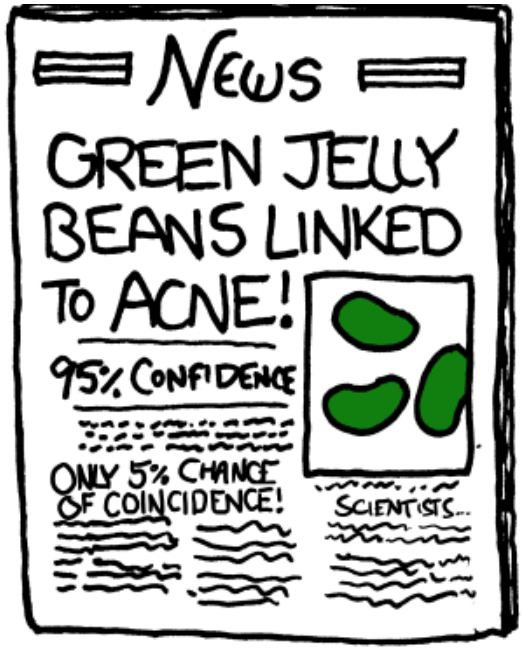


The curse of multiple testing

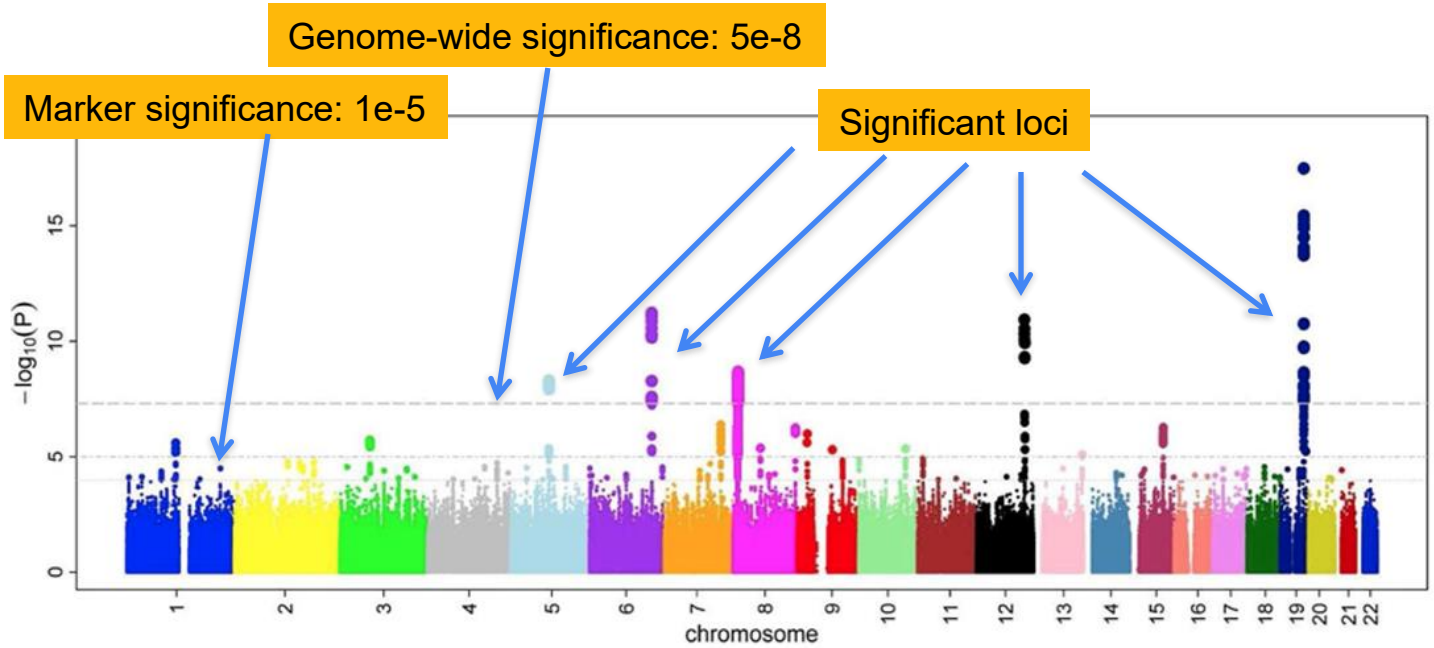


The curse of multiple testing





Manhattan Plot



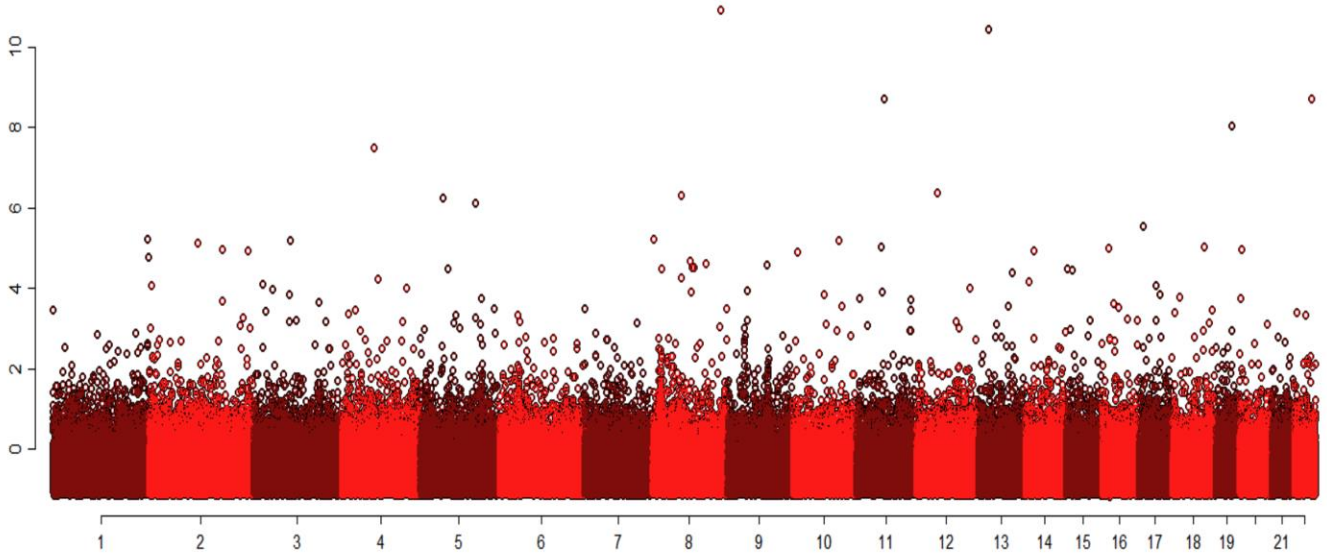
- The conventional genome-wide significance threshold is p-value ≤ 5×10^{-8}
- This threshold corresponds to a Bonferroni correction for approximately one million independent SNP tests ($0.05 / 10^6$).
- Nearby SNPs often form association peaks due to linkage disequilibrium (LD), as correlated variants tend to be inherited together.

Ikram et al, PLOS Genetics. 2010

Manhattan Plot

Genetic signatures of exceptional longevity in humans, Sebastiani et al. Science, 2010

Retracted July 2011 because of poor QC



- Highly significant SNPs appear isolated, with little support from nearby variants.
- Such signals may indicate genotyping errors or other QC artifacts and require further investigation.

First published GWAS

Complement Factor H Polymorphism in Age-Related Macular Degeneration

Robert J. Klein,[1] Caroline Zeiss,[2]* Emily Y. Chew,[3]* Jen-Yue Tsai,[4]* Richard S. Sackler,[1] Chad Haynes,[1] Alice K. Henning,[5] John Paul SanGiovanni,[3] Shrikant M. Mane,[6] Susan T. Mayne,[7] Michael B. Bracken,[7] Frederick L. Ferris,[3] Jurg Ott,[1] Colin Barnstable,[2] Josephine Hoh[7]†

Age-related macular degeneration (AMD) is a major cause of blindness in the elderly. We report a genome-wide screen of 96 cases and 50 controls for polymorphisms associated with AMD. Among 116,204 single-nucleotide polymorphisms genotyped, an intronic and common variant in the complement factor H gene (*CFH*) is strongly associated with AMD (nominal *P* value $<10^{-7}$). In individuals homozygous for the risk allele, the likelihood of AMD is increased by a factor of 7.4 (95% confidence interval 2.9 to 19). Resequencing revealed a polymorphism in linkage disequilibrium with the risk allele representing a tyrosine-histidine change at amino acid 402. This polymorphism is in a region of CFH that binds heparin and C-reactive protein. The *CFH* gene is located on chromosome 1 in a region repeatedly linked to AMD in family-based studies.

Age-related macular degeneration (AMD) is the leading cause of blindness in the developed world. Its incidence is increasing as the elderly population expands (*1*). AMD is characterized by progressive destruction of the retina's central region (macula), causing central field visual loss (*2*). A key feature of AMD is the formation of extracellular deposits called drusen concentrated in and around the macula behind the retina between the retinal pigment epithelium (RPE) and the choroid. To date, no therapy for this disease has proven to be broadly effective. Several risk factors have been linked to AMD, including age, smoking, and family history (*3*). Candidate-gene studies have not found any genetic differences that can account for a large proportion of the overall prevalence (*2*). Family-based whole-genome linkage scans have identified chromosomal regions that show evidence of linkage to AMD (*4–8*), but the linkage areas have not been resolved to any causative mutations.

Like many other chronic diseases, AMD is caused by a combination of genetic and environmental risk factors. Linkage studies are not as powerful as association studies for the identification of genes contributing to the risk for common, complex diseases (*9*). However, linkage studies have the advantage of searching the whole genome in an unbiased manner without presupposing the involvement of particular genes. Searching the whole genome in an association study requires typing 100,000 or more single-nucleotide polymorphisms (SNPs) (*10*). Because of these technical demands, only one whole-genome association study, on susceptibility to myocardial infarction, has been published to date (*11*).

Study design. We report a whole-genome case-control association study for genes involved in AMD. To maximize the chance of success, we chose clearly defined phenotypes for cases and controls. Case individuals exhibited at least some large drusen in a quantitative photographic assessment combined with evidence of sight-threatening AMD (geographic atrophy or neovascular AMD). Control individuals had either no or only a few small drusen. We analyzed our data using a statistically conservative approach to correct for the large number of SNPs tested, thereby guaranteeing that the probability of a false positive is no greater than our reported *P* values.

We used a subset of individuals who participated in the Age-Related Eye Disease Study (AREDS) (*12*). From the AREDS

[1]Laboratory of Statistical Genetics, Rockefeller University, 1230 York Avenue, New York, NY 10021, USA. [2]Department of Ophthalmology and Visual Science, Yale University School of Medicine, 330 Cedar Street, New Haven, CT 06520, USA. [3]National Eye Institute, Building 10, CRC, 10 Center Drive, Bethesda, MD 20892–1204, USA. [4]Biological Imaging Core, National Eye Institute, 9000 Rockville Pike, Bethesda, MD 20892, USA. [5]The EMMES Corporation, 401 North Washington Street, Suite 700, Rockville MD 20850, USA. [6]W. M. Keck Facility, Yale University, 300 George Street, Suite 201, New Haven, CT 06511, USA. [7]Department of Epidemiology and Public Health, Yale University School of Medicine, 60 College Street, New Haven, CT 06520, USA.

*These authors contributed equally to this work.
†To whom correspondence should be addressed. E-mail: josephine.hoh@yale.edu

First published GWAS

Complement Factor H Polymorphism in Age-Relate Macular Degeneration

Robert J. Klein,[1] Caroline Zeiss,[2*] Emily Y. Chew,[3*] Jen-Yue Tsai,[4*] Richard S. Sackler,[1] Chad Haynes,[1] Alice K. Henning,[5] John Paul SanGiovanni,[3] Shrikant M. M Susan T. Mayne,[7] Michael B. Bracken,[7] Frederick L. Ferri Jurg Ott,[1] Colin Barnstable,[2] Josephine Hoh[7†]

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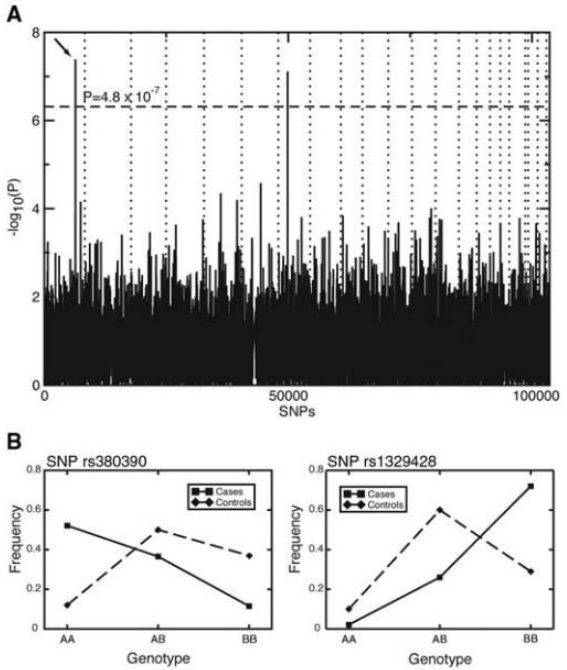


Fig. 1. (**A**) *P* values of genome-wide association scan for genes that affect the risk of developing AMD. $-\log_{10}(p)$ is plotted for each SNP in chromosomal order. The spacing between SNPs on the plot is uniform and does not reflect distances between SNPs on the chromosomes. The dotted horizontal line shows the cutoff for $P = 0.05$ after Bonferroni correction. The vertical dotted lines show chromosomal boundaries. The arrow indicates the peak for SNP rs380390, the most significant association, which was studied further. (**B**) Variation in genotype frequencies between cases and controls.

An early GWAS of 96 cases and 50 controls identified a strong association between variants in CFH and age-related macular degeneration (AMD).

ARTICLE

doi:10.1038/nature13595

Biological insights from 108 schizophrenia-associated genetic loci

Schizophrenia Working Group of the Psychiatric Genomics Consortium*

Schizophrenia is a highly heritable disorder. Genetic risk is conferred by a large number of alleles, including common alleles of small effect that might be detected by genome-wide association studies. Here we report a multi-stage schizophrenia genome-wide association study of up to 36,989 cases and 113,075 controls. We identify 128 independent associations spanning 108 conservatively defined loci that meet genome-wide significance, 83 of which have not been previously reported. Associations were enriched among genes expressed in brain, providing biological plausibility for the findings. Many findings have the potential to provide entirely new insights into aetiology, but associations at *DRD2* and several genes involved in glutamatergic neurotransmission highlight molecules of known and potential therapeutic relevance to schizophrenia, and are consistent with leading pathophysiological hypotheses. Independent of genes expressed in brain, associations were enriched among genes expressed in tissues that have important roles in immunity, providing support for the speculated link between the immune system and schizophrenia.

A large-scale GWAS involving up to 36,989 cases and 113,075 controls identified 108 schizophrenia-associated loci.

Nature 511, 421–427 (24 July 2014) doi:10.1038/nature13595

ARTICLE

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Biological in
schizophren

Schizophrenia Working Group of the P

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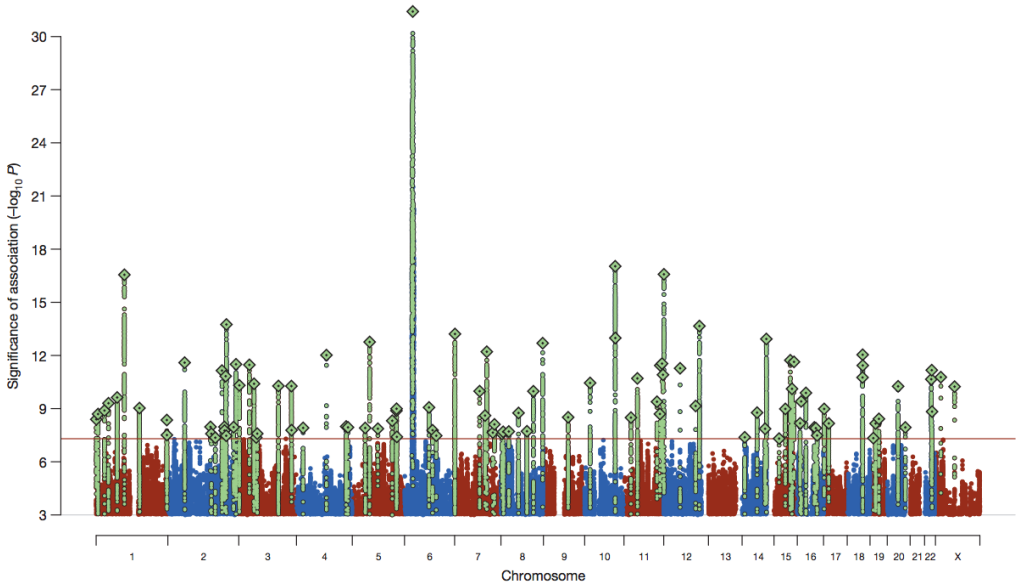
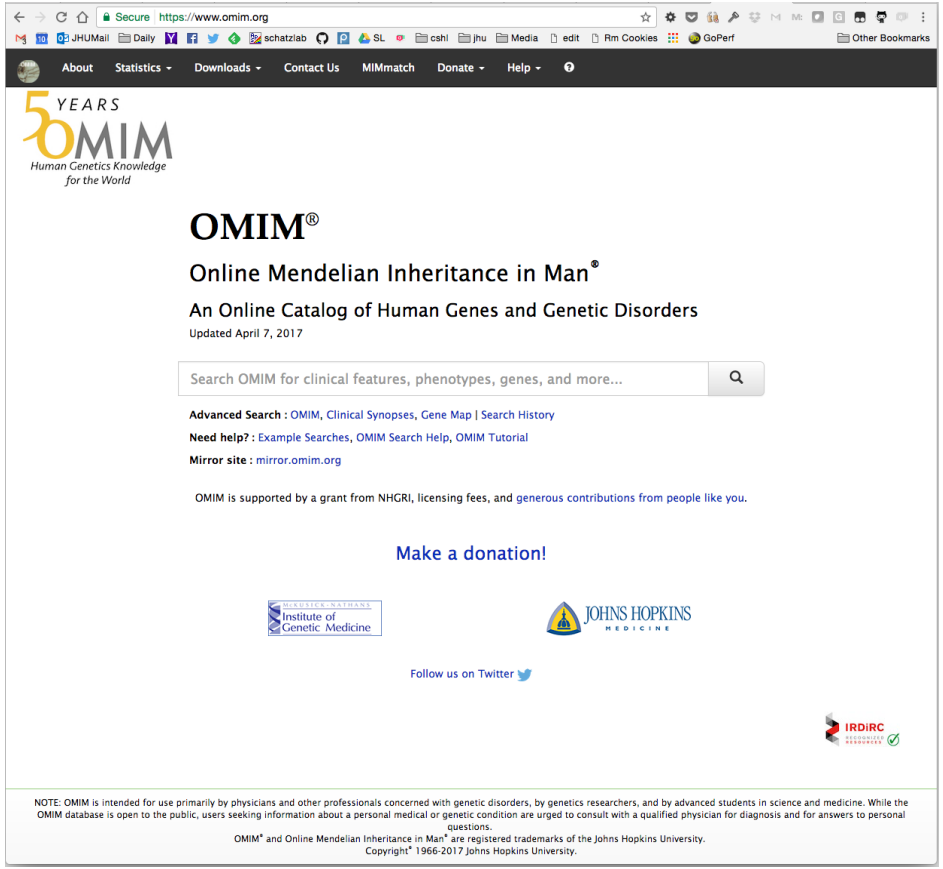


Figure 1 | Manhattan plot showing schizophrenia associations. Manhattan plot of the discovery genome-wide association meta-analysis of 49 case control samples (34,241 cases and 45,604 controls) and 3 family based association studies (1,235 parent affected-offspring trios). The *x* axis is chromosomal position and the *y* axis is the significance ($-\log_{10} P$; 2-tailed) of association derived by logistic regression. The red line shows the genome-wide significance level (5×10^{-8}). SNPs in green are in linkage disequilibrium with the index SNPs (diamonds) which represent independent genome-wide significant associations.

The associations were distributed across the genome, highlighting the polygenic architecture of schizophrenia.

Nature 511, 421–427 (24 July 2014) doi:10.1038/nature13595

OMIM



- For many different diseases and phenotypes, lists what are all of the known genetic associations
- Has records for nearly all genes, ~5k different conditions with known molecular basis, ~1k with unknown basis, ~1k with questionable basis
- Started at JHU 50 years ago ☺

GWAS Catalog

As of 2026-10-03, the GWAS Catalog contains 7845 publications,
1,198,394 top associations and 203,002 full summary statistics.



- Abdominal aortic aneurysm
- Acute lymphoblastic leukemia
- Adhesion molecules
- Adverse response to carbamapezine
- Adiponectin levels
- Age-related macular degeneration
- AIDS progression
- Alcohol dependence
- Alopecia areata
- Alzheimer disease
- Amyloid A levels
- Amyotrophic lateral sclerosis
- Angiotensin-converting enzyme activity
- Ankylosing spondylitis
- Arterial stiffness
- Asparagus anosmia
- Asthma
- Atherosclerosis in HIV
- Atrial fibrillation
- Attention deficit hyperactivity disorder
- Autism
- Basal cell cancer
- Behcet's disease
- Bipolar disorder
- Biliary atresia
- Bilirubin
- Bitter taste response
- Birth weight
- Bladder cancer
- Bleomycin sensitivity
- Blond or brown hair
- Blood pressure
- Blue or green eyes
- BMI, waist circumference
- Cleft lip/palate
- Cognitive function
- Conduct disorder
- Colorectal cancer
- Corneal thickness
- Coronary disease
- Creutzfeldt-Jakob disease
- Crohn's disease
- Cutaneous nevi
- Dermatitis
- Drug-induced liver injury
- Endometriosis
- Eosinophil count
- Eosinophilic esophagitis
- Erectile dysfunction and prostate cancer treatment
- Erythrocyte parameters
- Esophageal cancer
- Essential tremor
- Exfoliation glaucoma
- Eye color traits
- F cell distribution
- Fibrinogen levels
- Folate pathway vitamins
- Follicular lymphoma
- Fuch's corneal dystrophy
- Freckles and burning
- Gallstones
- Gastric cancer
- Glioma
- Glycemic traits
- Hair color
- Hair morphology
- Handedness in dyslexia
- HDL cholesterol

http://www.ebi.ac.uk/gwas/diagram

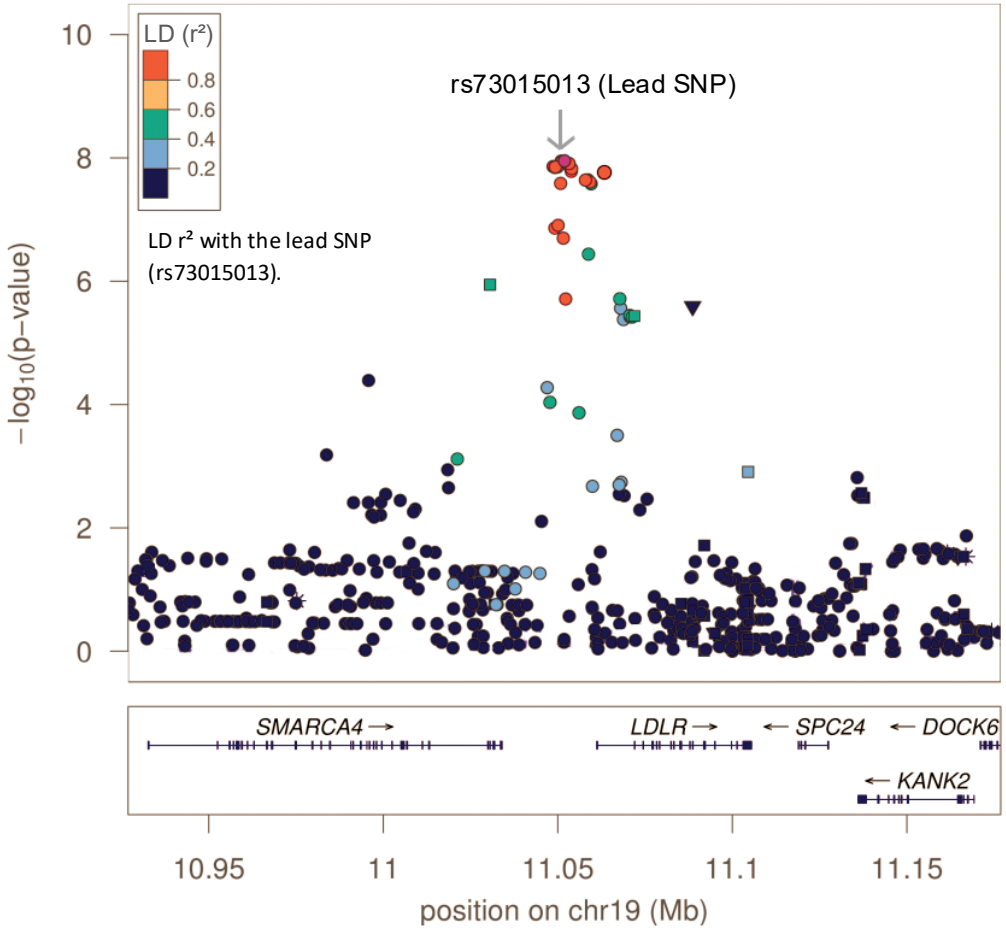
Genetic association studies and functional interpretation

1. How can we systematically identify variants associated with complex traits?
Genome-wide association studies (GWAS)

2. Among many associated variants, which are most likely causal?
Fine-mapping

3. Beyond trait associations, how does genetic variation relate to gene expression?
Expression quantitative trait loci (eQTL)

From GWAS association to causal variants



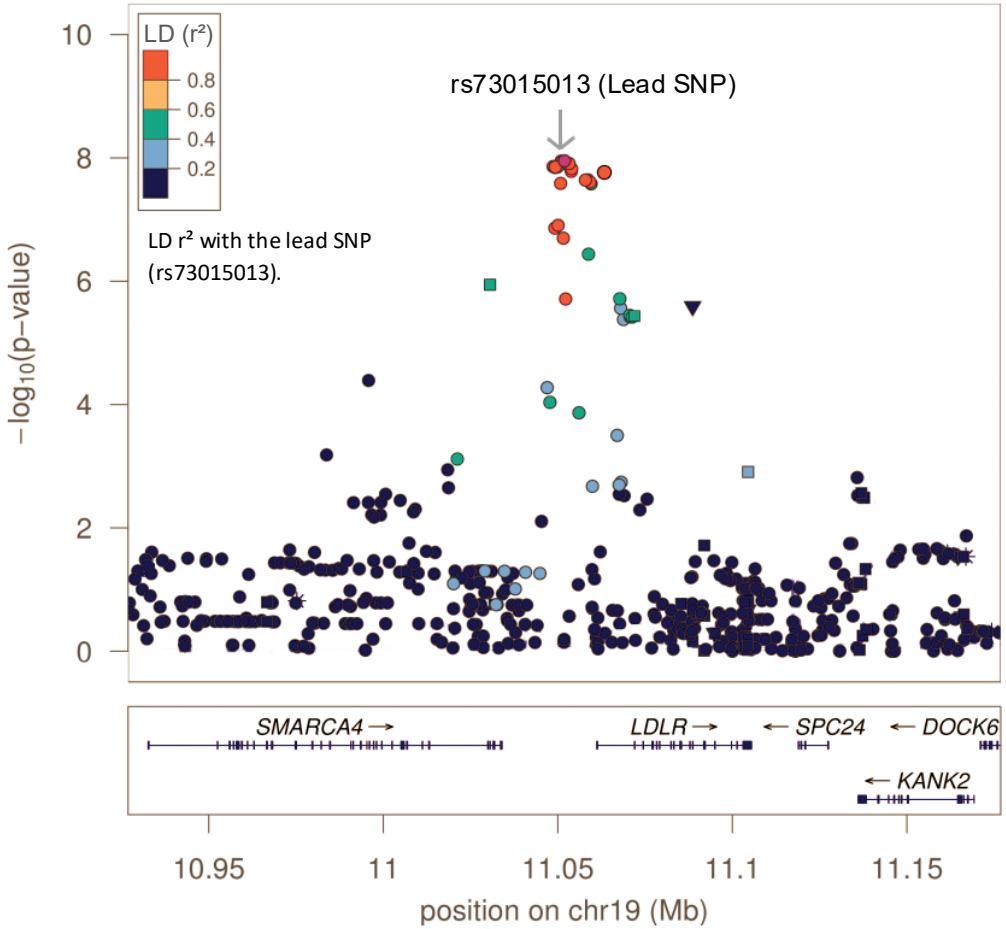
Linkage disequilibrium (LD)

- Nearby SNPs may be correlated and tend to be inherited together.
- Due to LD, a single association signal can produce a peak involving multiple SNPs.
- Colors indicate LD (r^2) with the lead SNP.
  - r^2 =1: perfect correlation
  - r^2 =0: no correlation

Identifying the causal variant

- The lead SNP (with the smallest P-value) is not necessarily the causal variant.
- Strong LD makes it difficult to distinguish causal variants from correlated SNPs.

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Which variants are most likely causal?

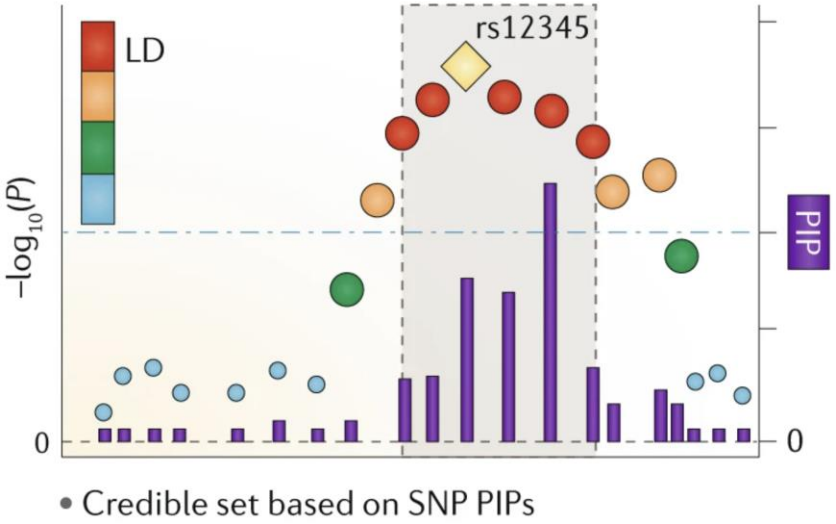
Fine-mapping analysis

Objective

- Prioritize variants most likely to drive a GWAS association signal.

Method

- Input: GWAS association statistics (effect sizes and standard errors) and LD correlations between variants.
- Evaluate: Compare possible causal variants or combinations of variants to determine which best explain the observed association signals.
- Estimate: Use Bayesian inference to calculate the probability that each variant has a causal effect.



Posterior inclusion probability (PIP)
The probability that a variant has a causal effect, given the data and model.

95% credible set
A set of variants with 95% posterior probability of containing a causal variant for a given signal.

Schaid et al., Nature Reviews Genetics, 2018

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Expression quantitative trait loci (eQTLs)

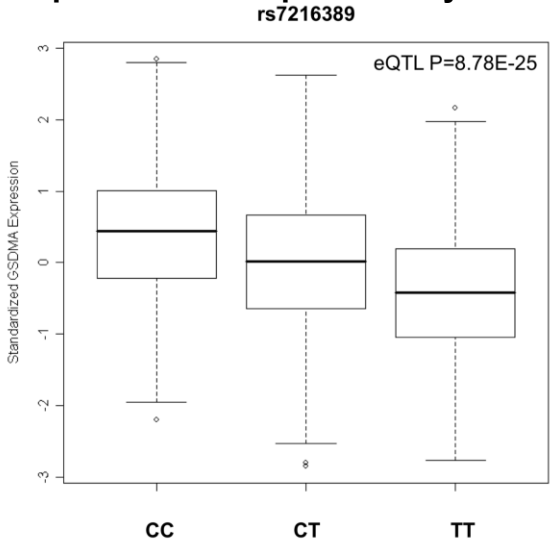
Objective

- Identify genetic variants associated with differences in gene expression.

Required data from the same individuals

- Genotypes: effect-allele dosage (0, 1, or 2 copies)
- Gene expression: normalized expression levels measured for each individual
- Covariates: sex, ancestry PCs, and technical factors (e.g., batch or PEER factors [inferred factors capturing hidden sources of expression variation])

Example: GSDMA expression by rs7216389 genotype (lung)



- Expression differs across rs7216389 genotypes (CC, CT, and TT).
- Higher T-allele dosage is associated with lower GSDMA expression.

Cis-eQTL mapping

Objective

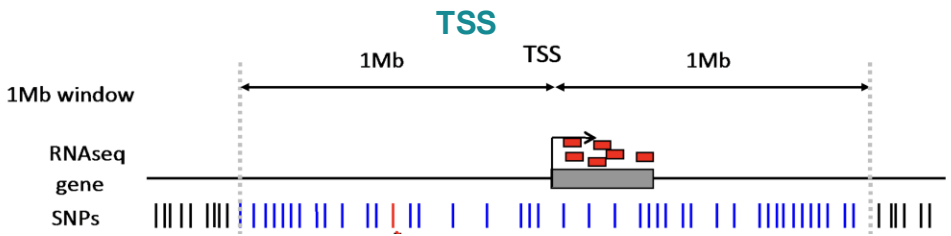
- For each gene, test nearby variants for association with its expression level.

Method

- Define a cis window around each gene (commonly ±1 Mb of the transcription start site).
- Fit a linear regression model relating genotype dosage to expression, adjusting for covariates.
- Test the association for each variant-gene pair and correct for multiple testing.

Cis window

Variants within ±1 Mb of the TSS



Additive linear regression

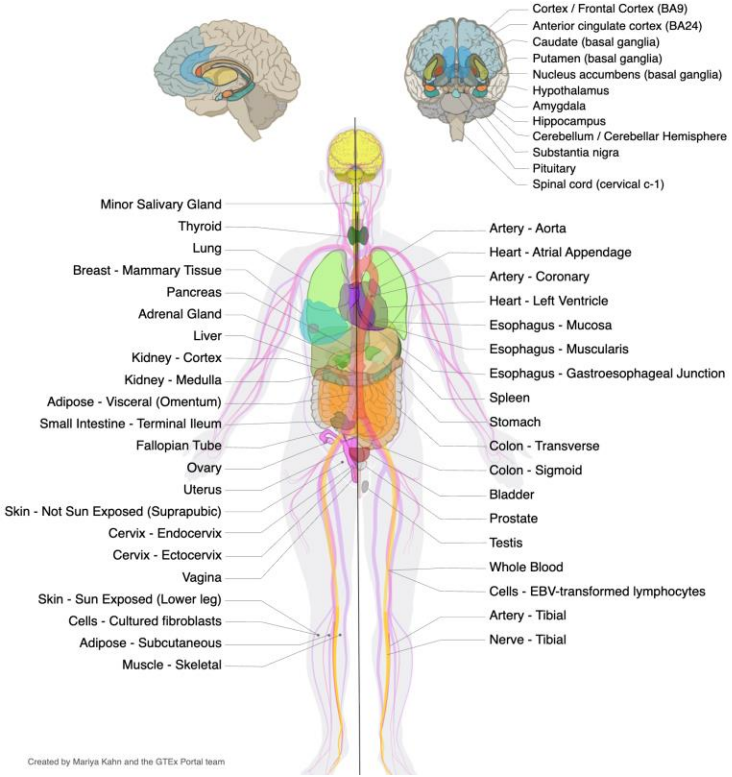
$$Y_i = \alpha + \beta G_i + \gamma^T X_i + \varepsilon_i$$

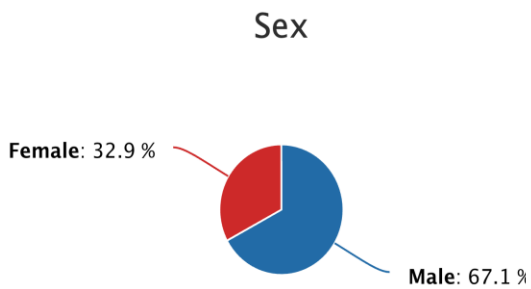
Y_i	Normalized expression of a gene
G_i	Effect-allele dosage (0, 1, or 2)
X_i	Covariates (e.g., sex, PCs, PEER)
β	Expression change per effect allele

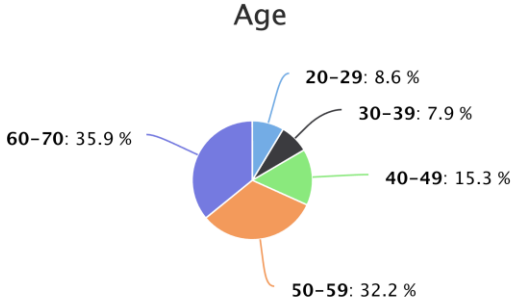
Test H_0: $\beta = 0$ Control FDR across genes

Dermitzakis M. GTEx Pilot Analysis, 2015

Genotype-Tissue Expression (GTEx)







V10 Release	# Tissues	# Donors	# Samples
Total	54	946	19788
With Genotype	54	946	19788
Has eQTL Analysis*	50	943	19466

* Number of samples with genotype >= 70

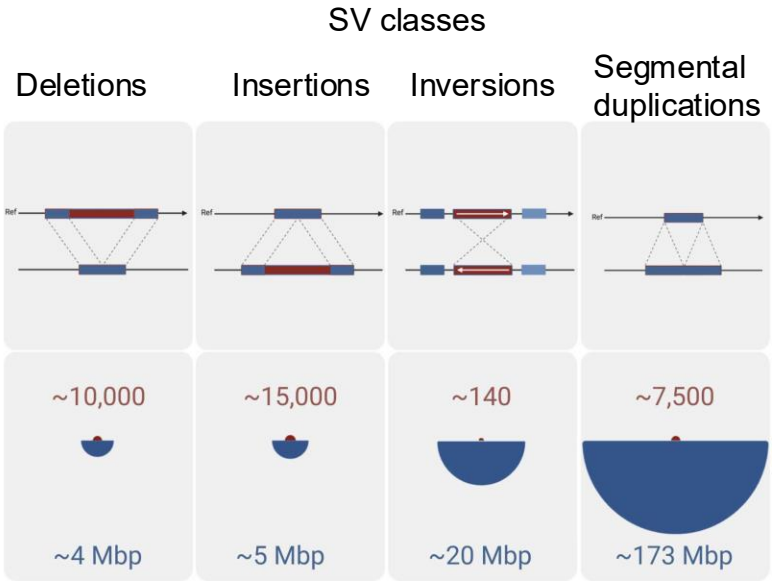
https://gtexportal.org/home/

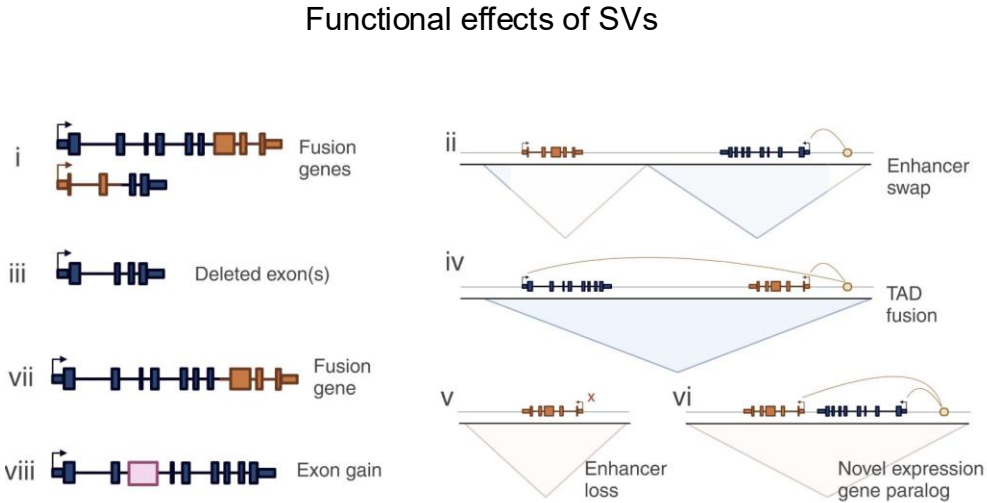


Outline

- Early discoveries in human disease genetics
- Genetic association studies and functional interpretation
- Population-scale long-read genomics in All of Us

Why structural variants matter?



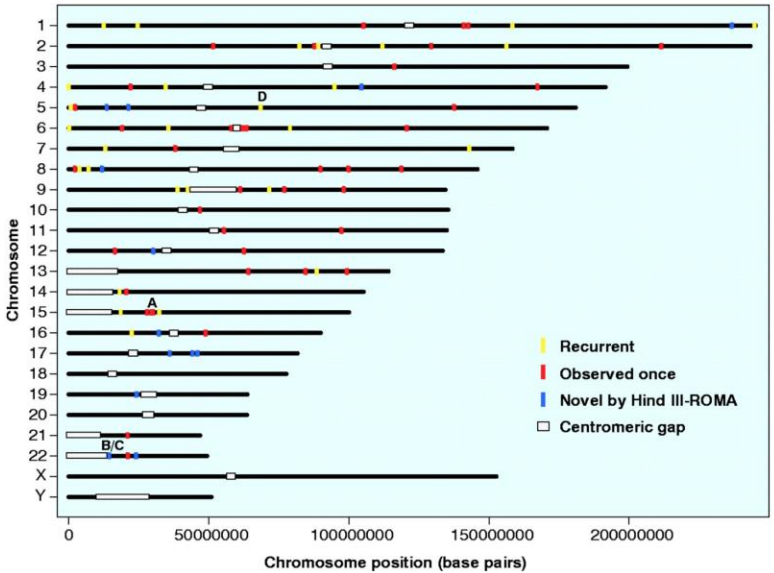


Structural variants (SVs) affect large genomic regions and have the potential to alter gene function and regulation.

Karageorgiou et al., Curr Opin Genet Dev, 2024

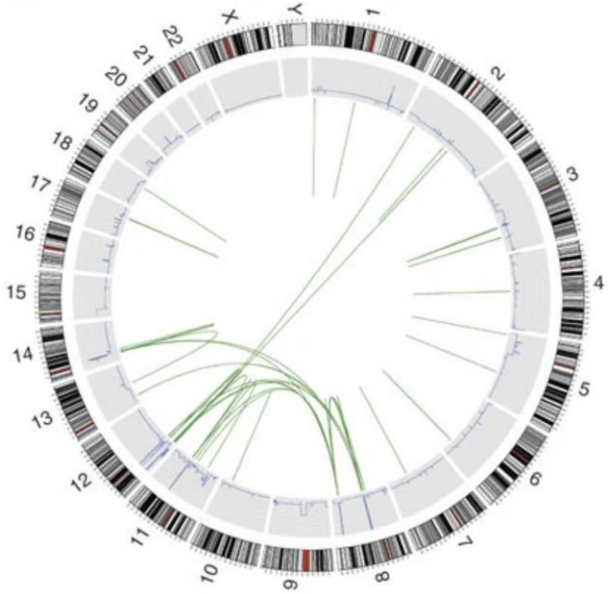
Early evidence for the clinical relevance of structural variation

Genome-wide map of copy-number polymorphisms (CNPs), including loci associated with disorders



A-D mark regions associated with Prader–Willi/Angelman, cat eye, DiGeorge/velocardiofacial syndromes, and spinal muscular atrophy.

Genome-wide acquired rearrangements NCI-H2171 small-cell lung cancer cell line

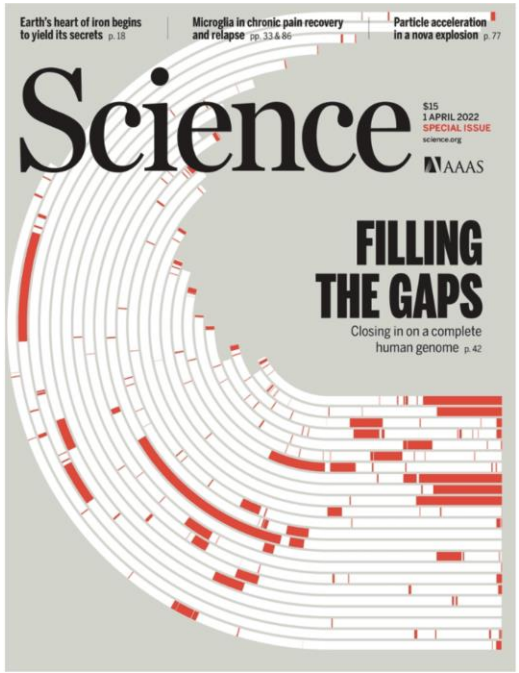


Outer ring: chromosomes; blue: copy number; green links: somatic rearrangements.

Sebat et al., Science, 2004; Campbell et al., Nat Genet, 2008

Long reads are reshaping human reference genomes

Telomere-to-Telomere (T2T) project



- Resolves previously inaccessible regions, including centromeres and telomeres
- Provides the first complete T2T human genome assembly

Human Pangenome Reference Consortium (HPRC)



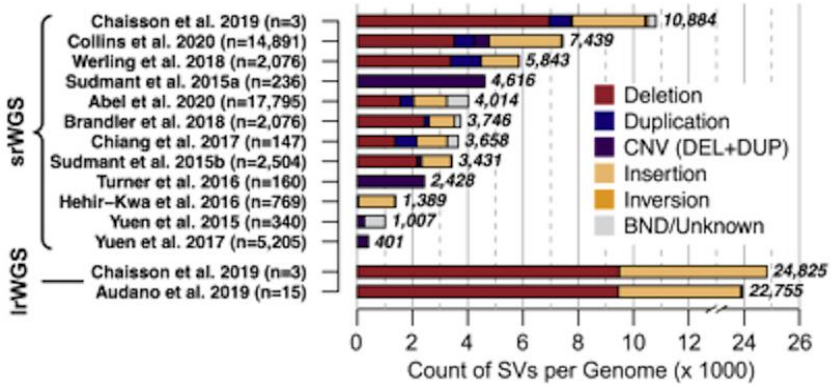
- Captures global human genomic diversity
- Represents multiple haplotypes in a graph-based reference

Liao et al., Nature, 2023; Nurk et al., Science, 2022

Long reads improve structural variant discovery

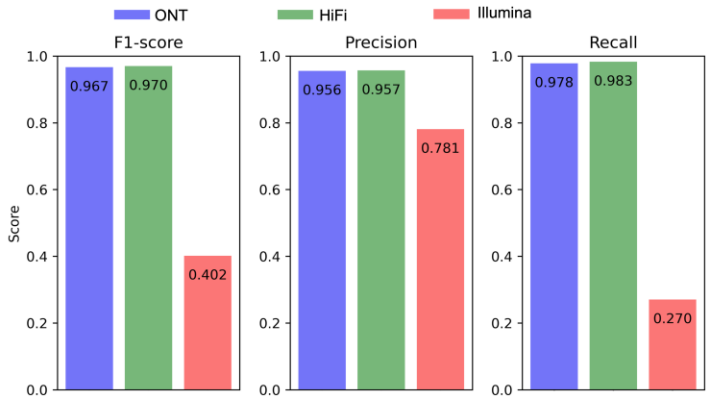
- Detects more SVs across the genome, with improved sensitivity across SV types
- Higher precision and recall compared to short-read approaches

Comparison of SV callsets from srWGS and



srWGS: short-read whole-genome sequencing
lrWGS: long-read whole-genome sequencing

GIAB HG002 benchmark



Zhao et al. American journal of human genetics, 2021; Li et al., Genome Res, 2025

All of Us Research Program: Advancing Precision Medicine

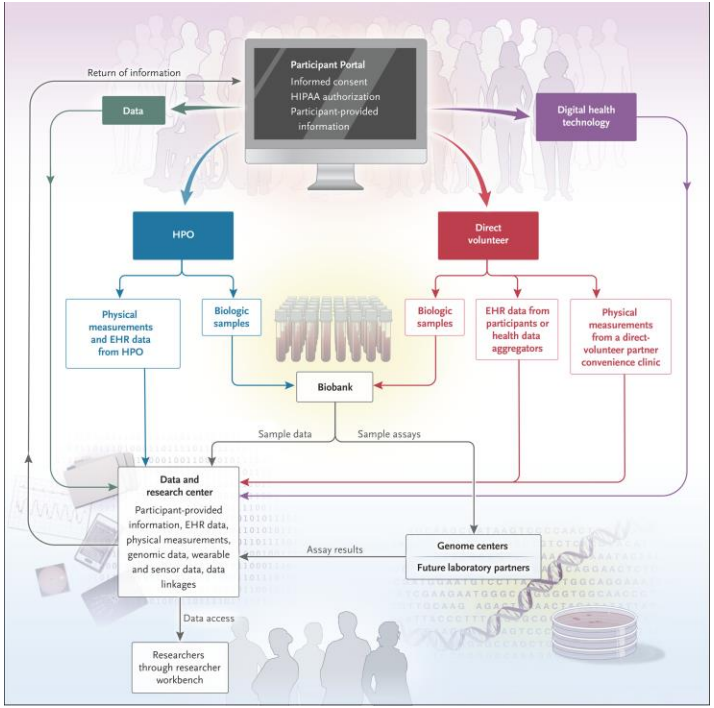
Why start *All of Us*?

- Genomic, Electronic Health Records (EHR), and other personalized data for 1M US citizens
- One-size-fits-all care often falls short for individual patients
- Fragmented health records limit coordinated, holistic care
- Redundant data collection slows discovery and increases cost

What we aim to achieve?

- Identify risk factors for both common and complex diseases
- Understand which treatments work best for people of different backgrounds
- Match participants to clinical studies that fit their needs
- Explore how technology can support personalized health

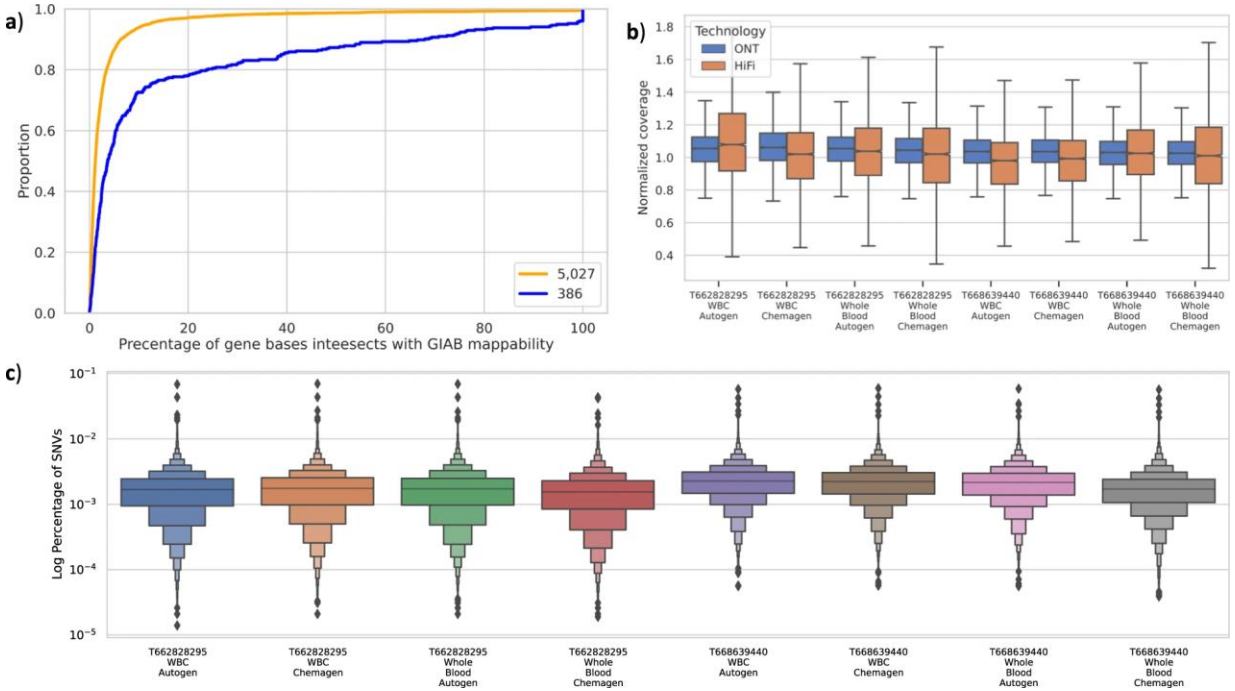
From promise to progress --
Shaping the future of health research



The "All of Us" research program
All of Us Research Program Investigators (2019) *NEJM*
doi: 10.1056/NEJMsr1809937

Long-read sequencing for *All of Us*

A pilot study of a few cell lines



Access to more of the genome and more complex variations:

- Structural variations, tandem repeat, segmental duplications and other complex variations
- Variations in repetitive sequences
- Improved variant identification, including within medically relevant genes
- Greatly improved phasing

Utility of long-read sequencing for *All of Us*
Mahmoud et al (2024) Nature Communication.

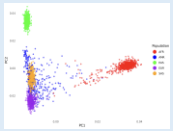
Long-read sequencing and variant discovery in 1,027 All of Us participants self-identified as Black or African American



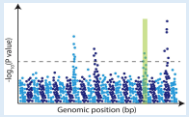
Data Overview: Genomic and EHR



Clinical and Biomedical Variations



Cross-Population eQTLs



Genome-Wide Analysis of Disease Associations



https://www.pexels.com/photo/hands-with-colorful-traditional-beads-bracelets-8628442/

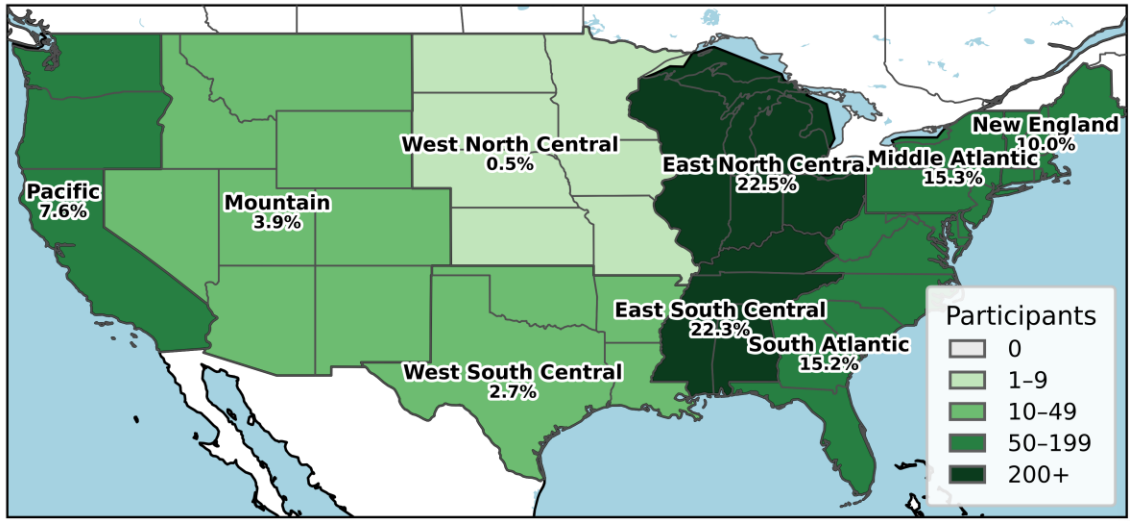
All of Us Long Read Sequencing Participants

Sample information

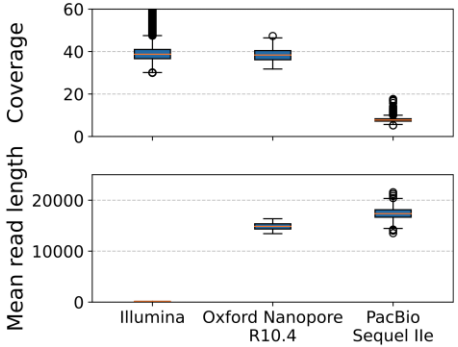
- 1,027 self-identified Black or African American participants
- Selected from the All of Us Research Program (CDR v7), which includes 245,388 short-read participants across the United States



Geographic sampling locations for 1,027 long read samples



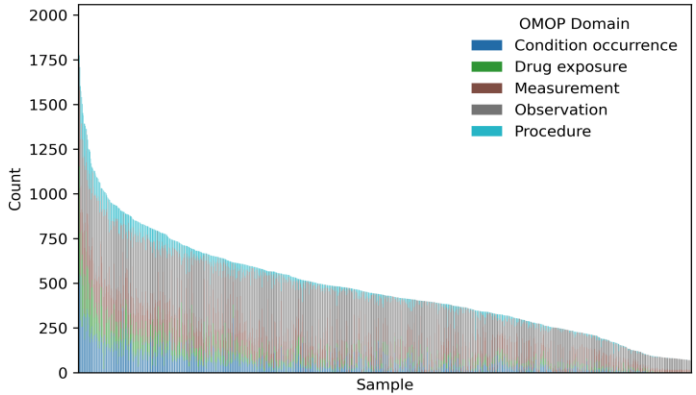
Sequencing Performance



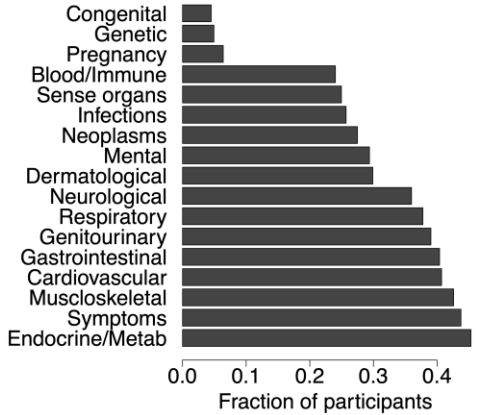
- 1027 samples sequenced PacBio HiFi: ~8× coverage, mean read length: 17.5 kbp
- Long-reads were 115× longer than Illumina short-reads.
- 50 samples additionally sequenced with ONT (~35× coverage) for validation.

Electronic Health Records

- Clinical data mapped to five domains of the OMOP Common Data Model (CDM)



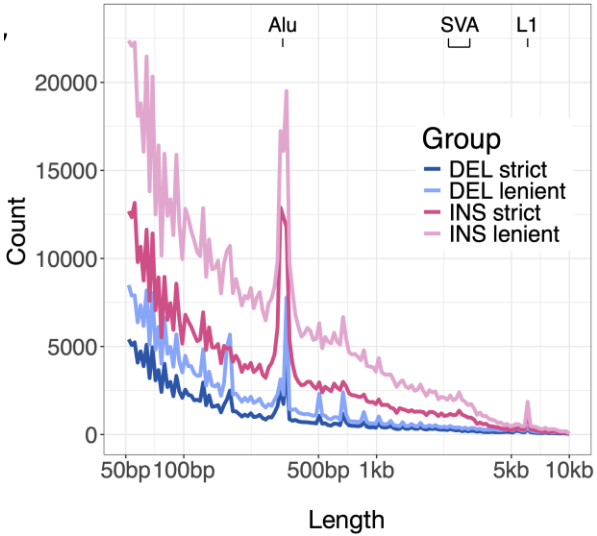
- Disease status among the 1,027 AoU long-read participants



Variant Discovery

Sample: 1,027 long-read *AoU* samples + 47 HPRC samples, sequenced at 8× depth.

Distribution of SV lengths for insertions and deletions



Tools
- PAV, Sniffles2, and PBSV

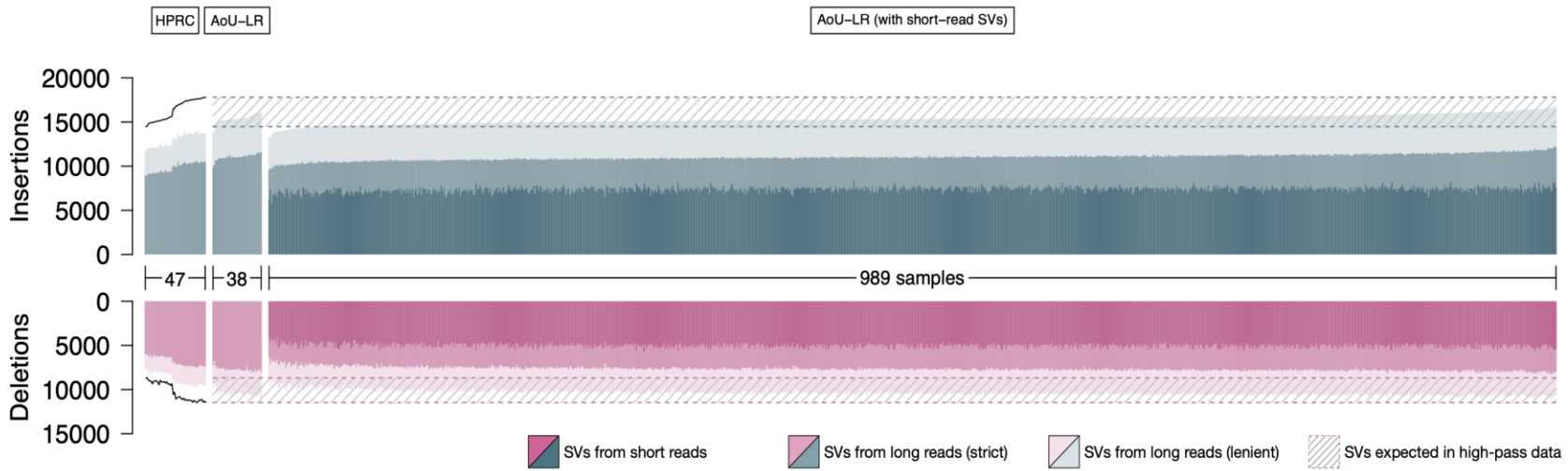
Process
- Applied Truvari for per-sample clustering and normalization.
- Trained a per-sample XGBoost classifier trained to remove false positives with Kanpig
- Generated cohort-wide SV callsets by merging filtered per-sample callsets using Truvari

- **Lenient callset**: 1,213,876 SVs - prioritized for sensitivity
- **Strict callset**: 665,869 SVs - prioritized for high specificity

Variant Discovery

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Comparison of short-read and long-read SV callsets



Long-read sequencing detected substantially more SVs than short-read sequencing

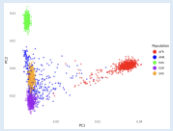
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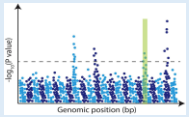
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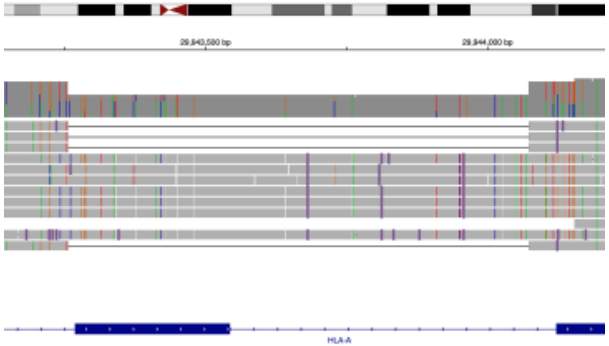
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SVs in Medically Relevant and Complex Genomic Regions

key gene sets intersecting with AoU SVs

	OMIM genes [1]	Genes associated with inherited diseases [2]	Genes associated with cancer [3]	American College of Medical Genetics and Genomics [4]
Intersecting gene body	10,968 (62.9%)	3,565 (70.9%)	609 (80.3%)	70 (86.4%)
Intersecting gene exons	2,827 (16.2%)	913 (18.2%)	152 (20.1%)	11 (13.6%)

- 723 LRS-unique SVs within the major histocompatibility complex (MHC), a repetitive and highly diverse genomic regions.
- An 816 bp deletion intersecting key immune-related *HLA-A*, resolved exclusively by long-read sequencing.



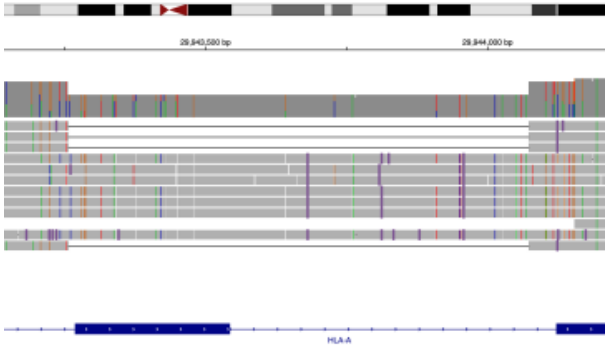
[1] https://www.omim.org/
[2] Wagner J, Olson ND, Harris L, et al. Curated variation benchmarks for challenging medically relevant autosomal genes. *Nat Biotechnol*. 2022;40(5):672-680
[3] https://www.oncokb.org/cancer-genes
[4] Miller DT, Lee K, Abul-Husn NS, et al. ACMG SF v3.2 list for reporting of secondary findings in clinical exome and genome sequencing: A policy statement of the American College of Medical Genetics and Genomics (ACMG). *Genet Med*. 2023;25(8):100866.

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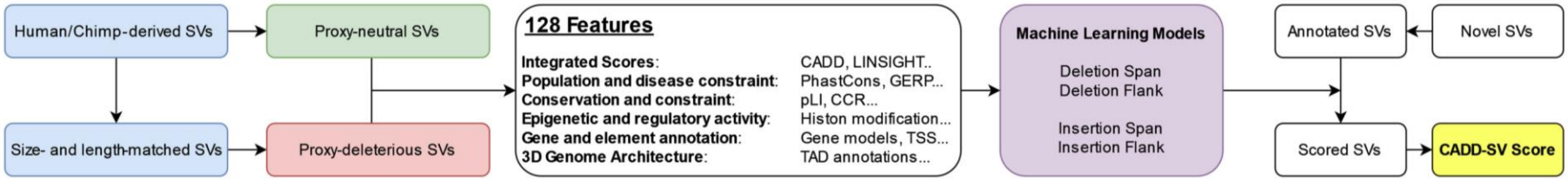
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CADD-SV: Predicting SV Deleteriousness

Annotations
128 genomic features covering conservation, gene annotations, regulatory activity, and 3D genome organization, summarized across SV spans and flanks.

Model
Random forests trained on fixed human/chimpanzee-derived SVs (proxy-neutral) and matched simulated SVs (proxy-deleterious).

Scoring
The maximum span/flank model score is converted to a PHRED-scaled score relative to gnomAD SVs of the same type.



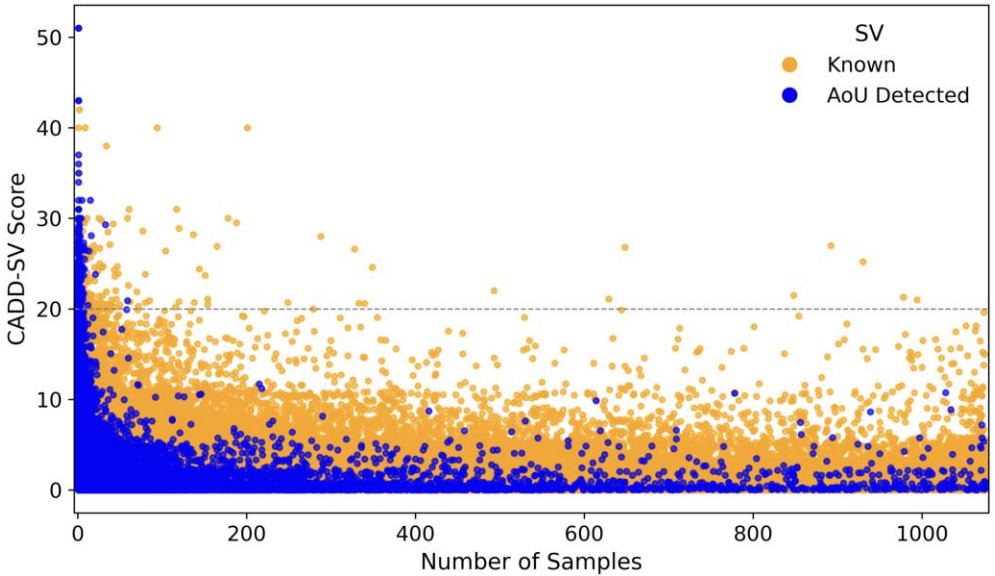
PHRED-scaled score: 10 = top 10%; 20 = top 1%; 30 = top 0.1% of reference SVs.

Higher scores indicate greater predicted deleteriousness

Kleinert & Kircher, Genome Research, 2022

Evaluation of Structural Variant Impact

- CADD-SV scores structural variants to prioritize those with potentially pathogenic impact for genomic studies.



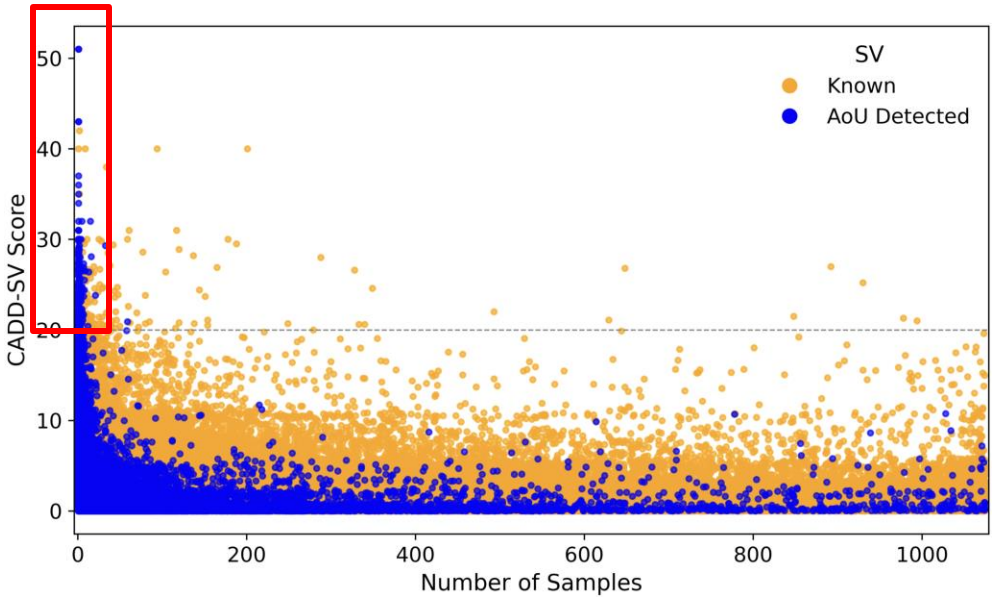
"Known" SVs are those found in at least one of the 1000G-ONT, HGSVC, or HPRC datasets
"AoU Detected" SVs are those absent in the dataset

- 473 SVs were identified as likely pathogenic (score ≥20), which are rare in the AoU cohort.
- SVs with scores ≥30 (pathogenic) are particularly rare, suggesting purifying selection.
- Among the likely pathogenic SVs absent from these external resources (n=273):
  - 172 SVs overlapped with 170 medically relevant genes
  - 73 SVs intersected with 69 high-priority disease genes
  - 15 SVs impacted 14 cancer risk-associated genes
- SVs absent from long-read population datasets showed lower allele frequencies in the AoU cohort compared to shared variants.

Kleinert & Kircher, Genome Res. 2022; Amberger et al., Nucleic Acids Res. 2019; Wagner et al., Nat Biotechnol. 2022; Chakravarty et al., JCO Precis Oncol. 2017

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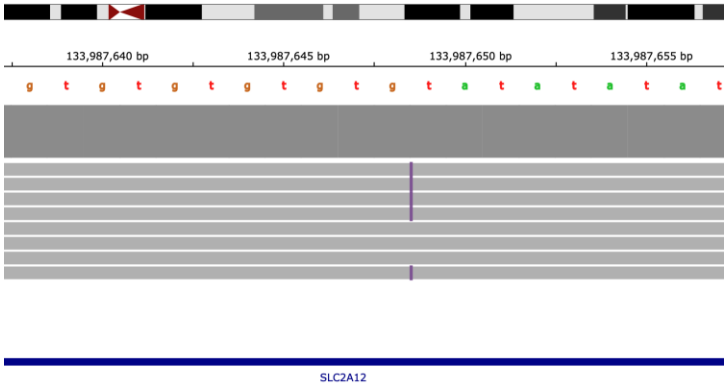
Evaluation of Structural Variant Impact

Objective

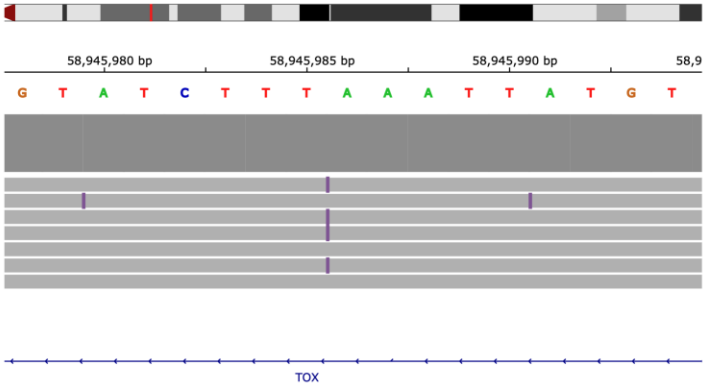
- Leverage available EHR data to explore the potential clinical consequences of high-impact SVs.

Examples of high-impact SVs

- A 52 bp insertion (CADD-SV score: 51), intersected the 3' UTR of *SLC2A12*, which encodes the glucose transporter GLUT12.
- The carrier had documented Type 2 diabetes and hyperglycemia, suggesting a metabolic impact.



- A 536 bp insertion (CADD-SV score:43), intersected an intron of *TOX*, a regulator of asthma pathogenesis.
- The individual carrying this variant had an asthma diagnosis, supporting the potential functional relevance of this SV.



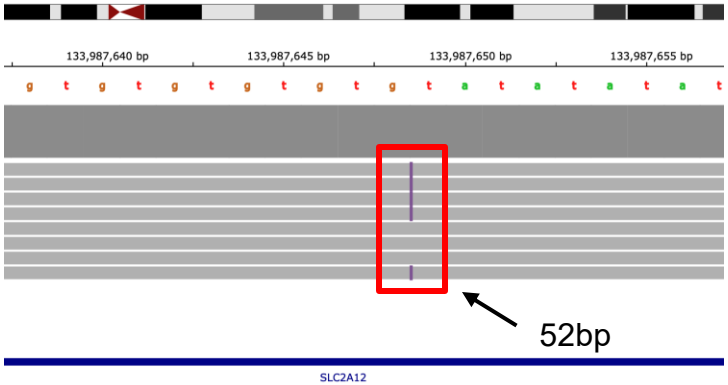
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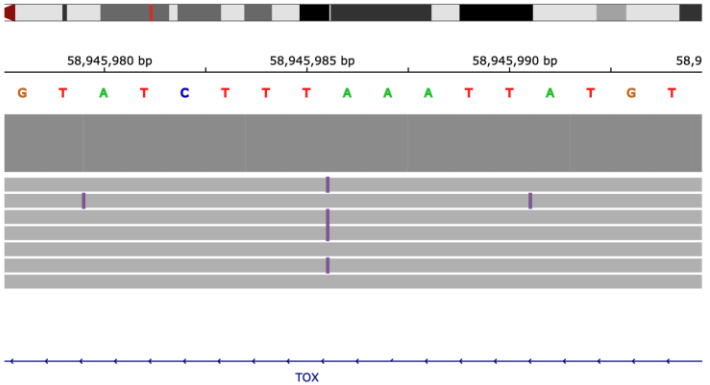
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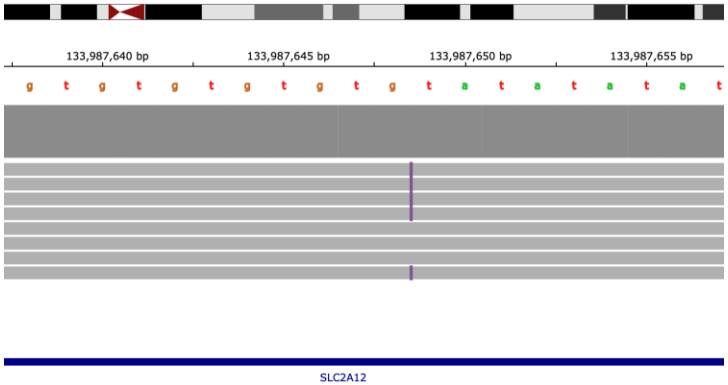
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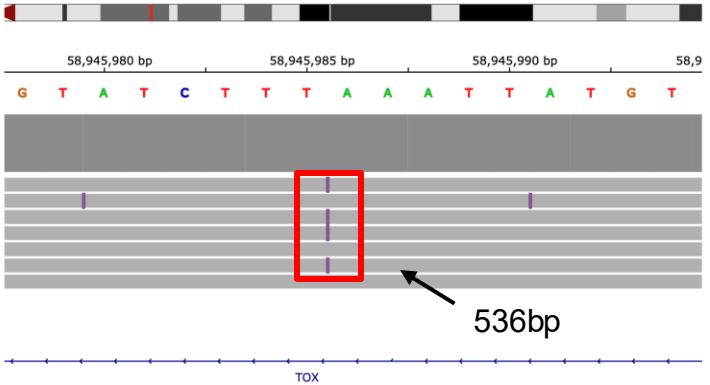
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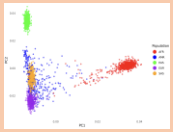
Long-read sequencing and variant discovery in 1,027 All of Us participants self-identified as Black or African American



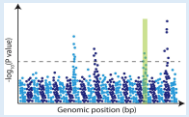
Data Overview: Genomic and EHR



Clinical and Biomedical Variations



Cross-Population eQTLs



Genome-Wide Analysis of Disease Associations

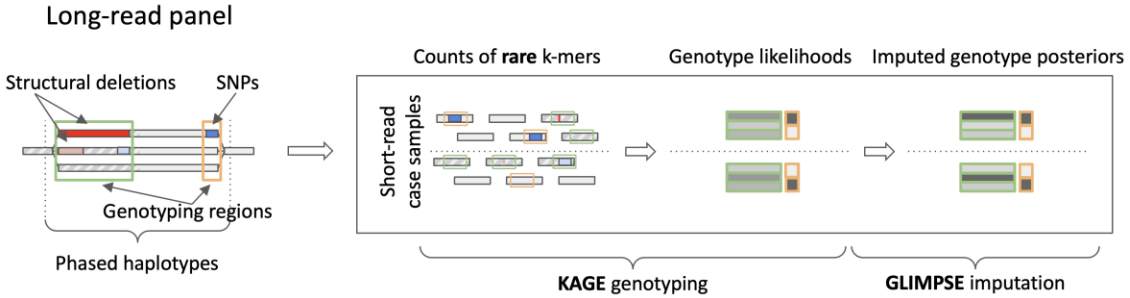


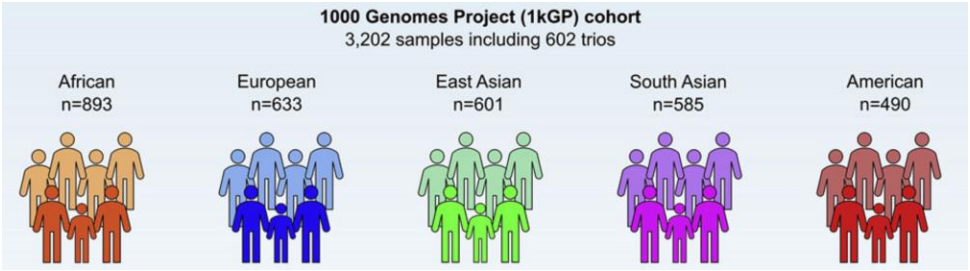
https://www.pexels.com/photo/hands-with-colorful-traditional-beads-bracelets-8628442/

Genotyping and Imputation of AoU Variants with 3,202 1KGP Samples

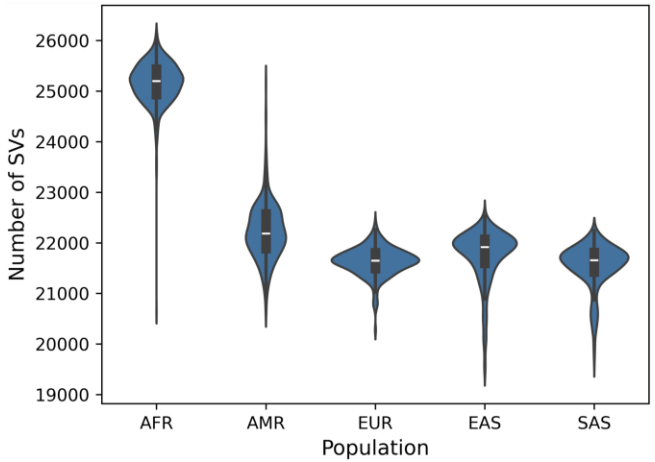
Aim:

- Interpret population-level structural variant patterns in the AoU dataset using data from 1KGP cohorts.





Number of imputed AoU SVs per individual



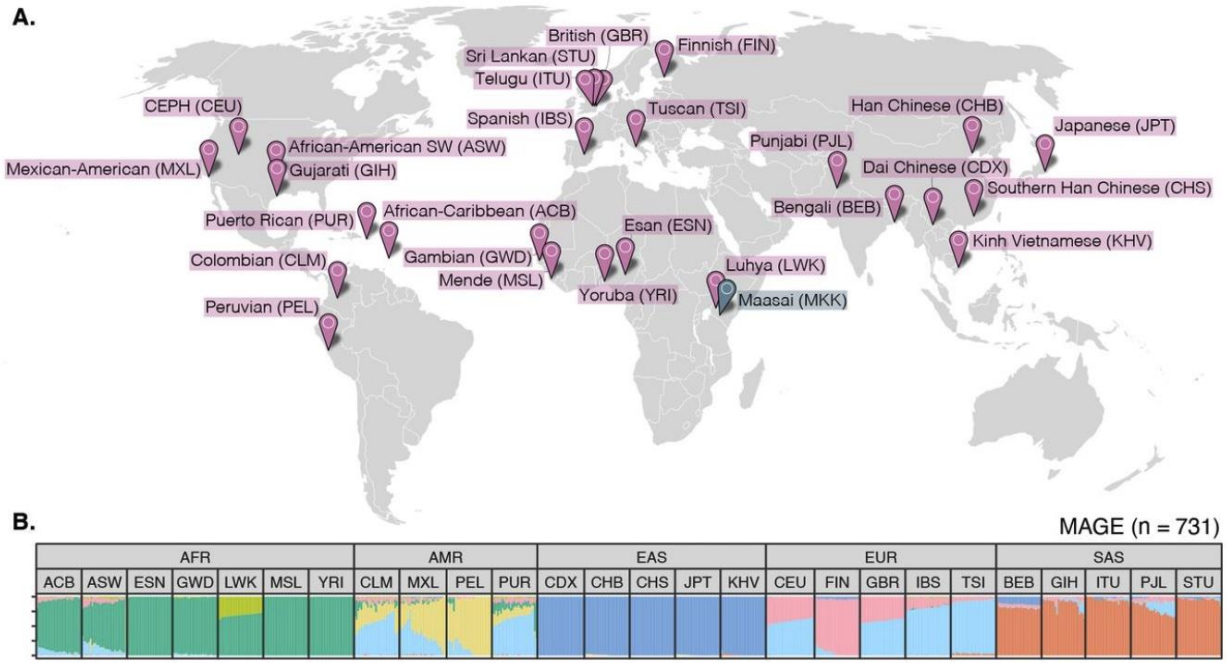
Imputation accuracy over 90% (outside of tandem repeats) using HPRC as the ground truth

Byrska-Bishop et al., Cell, 2022

1000 Genomes & MAGE Datasets

Components

- Matched DNA-seq (1000 Genomes) and RNA-seq (MAGE) data from 731 individuals, representing 26 globally distributed populations across five continents.

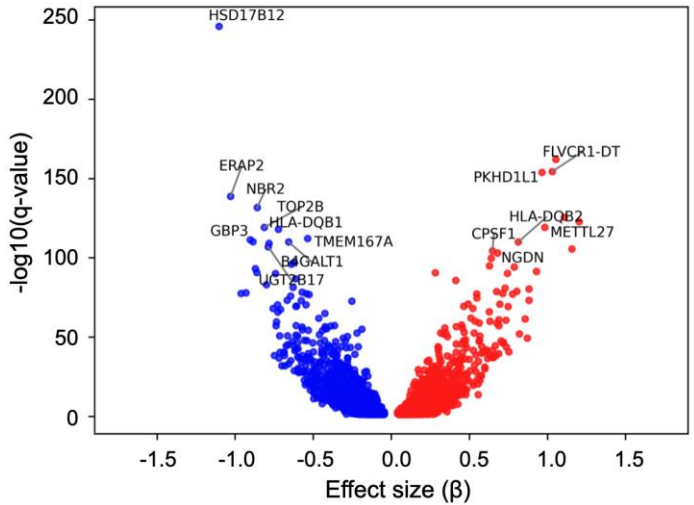


Taylor et al., Nature, 2024

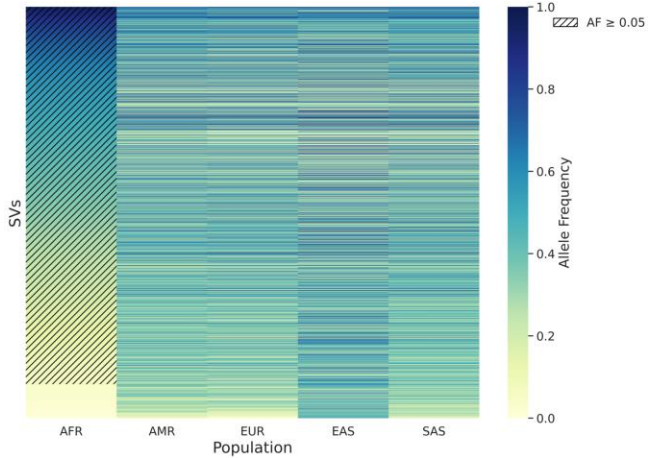
SV-eQTL Analysis

- Identified 3,654 significant SV-gene pairs, involving 2,582medically relevant OMIM genes, 718 high-priority disease genes, and 183 cancer-associated genes.
- 626 eGenes not reported in prior long-read SV–]eQTL studies (31 diverse long-read genomes[1] and 65 samples[2] from the 1KGP-ONT dataset)
- Most (90%) SVs in significant SV-gene pairs were common in all populations, allowing reliable assessment of their impact on gene expression.

Relationship between the best-fit slope and q-values



Allele frequency of SVs in SV-eQTLs



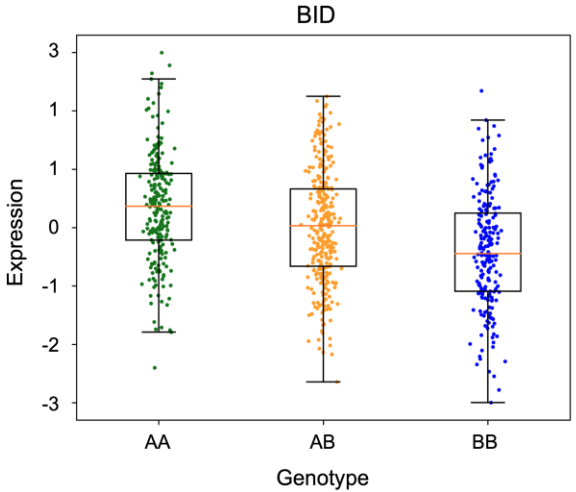
Rows represent SV AF across five populations."///" marks SVs common in Africans.

[1] Gustafson et al., Genome Res, 2024
[2] Kirsche et al., Nat Methods, 2023

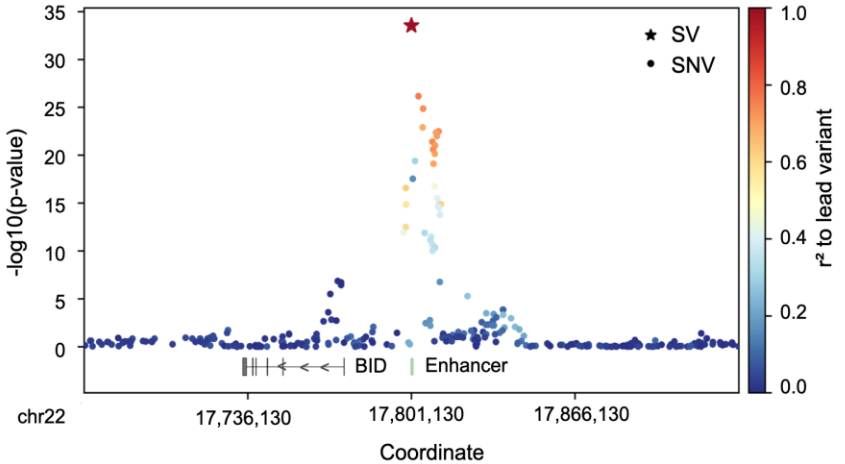
SV-eQTL Analysis

- In 120 eGenes, SVs are the top CAVIAR-predicted causal variants compared with SNVs within a 1 Mbp window

Effect of deletion on BID
(q-value = 4.41×10^{-31})



Manhattan plot of the DEL and nearby SNVs
(SV PIP = 1; highest PIP at the locus)



- A 322 bp deletion within an enhancer upstream of *BID*, a pro-apoptotic BCL-2 family member involved in cancer progression and immune regulation, was identified as the top CAVIAR-predicted causal variant.

- Disruption of enhancer activity reduces BID expression, indicating the role of regulatory elements in gene regulation.

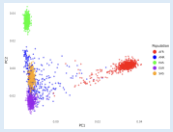
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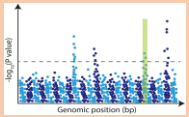
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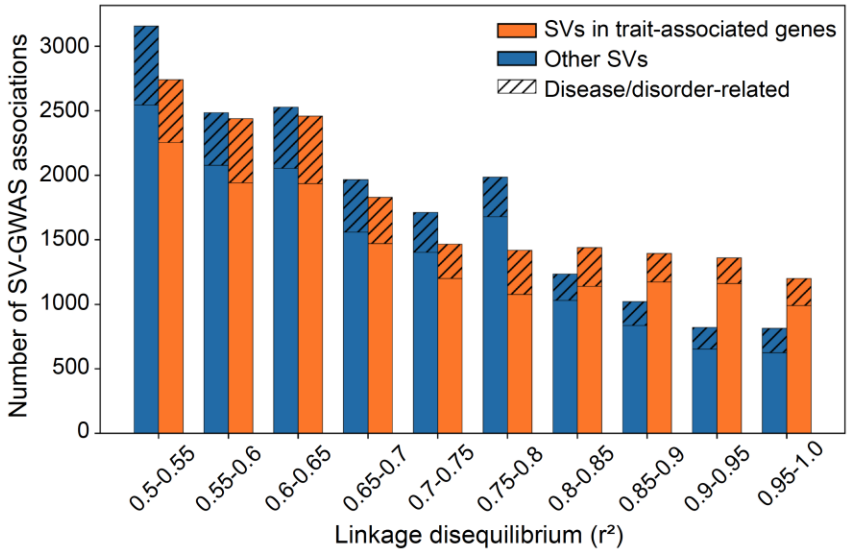
Structural Variation in Linkage Disequilibrium with GWAS-Significant SNVs

Objective

- Identify structural variants that may be relevant to traits or diseases associated with GWAS signals.

Method

- Calculated linkage disequilibrium (LD) between our SVs and SNPs from NHGRI-EB GWAS Catalog
- Prioritized SV-SNV pairs with genome-wide significance (p-value < 5×10^{-8}) and LD ($r^2 \geq 0.5$).



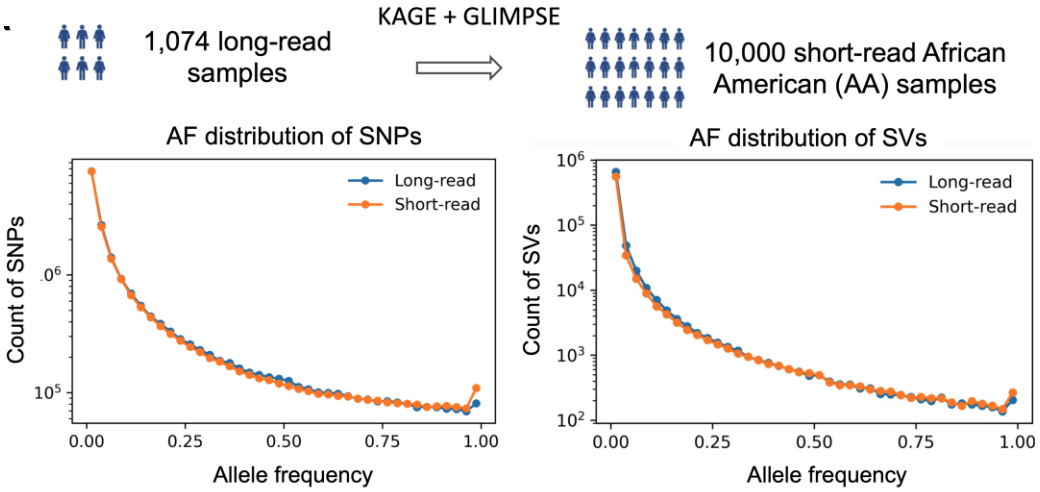
- 6,019 traits were significantly associated with SVs, including 1,103 related to diseases or disorders.
- Among these, 537 medically relevant traits had SVs in high LD ($r^2 \geq 0.8$) with SNVs.

Phenome-wide SV Analysis Identifies Disease Risk Factors

Objective

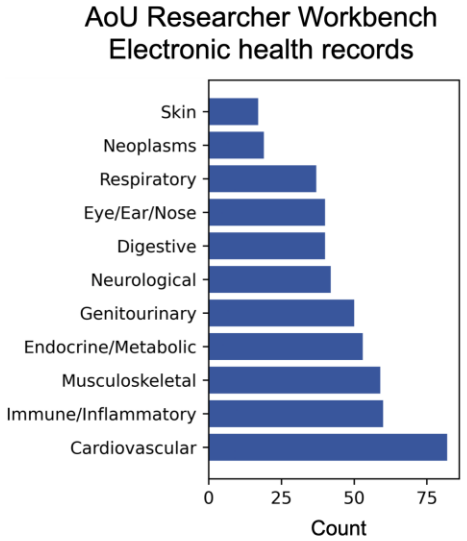
- Evaluate the clinical impact of SVs by integrating genomic data with electronic health records from an expanded cohort.

Genotyping and imputation with the long-read panel



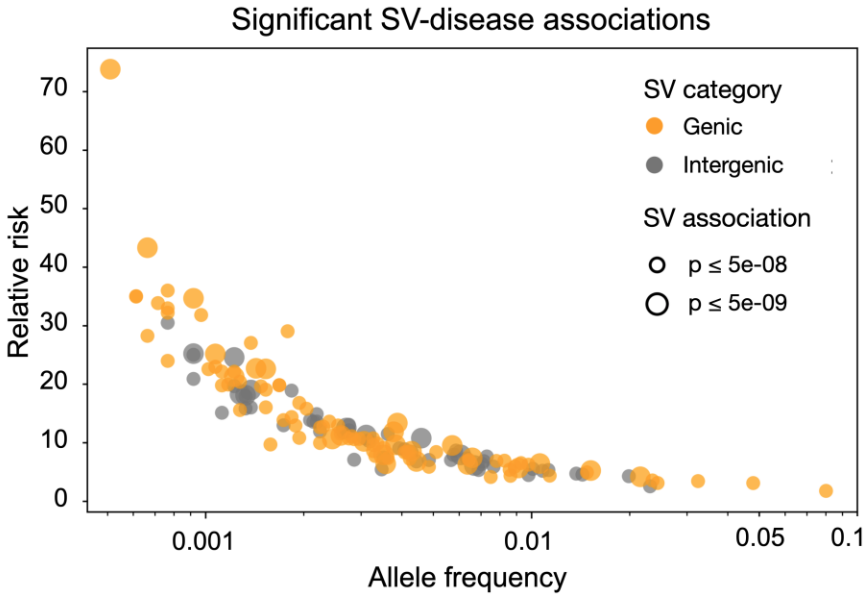
18.8M SNPs and 773K SVs identified in 1,074 long-read AA samples
18.5M SNPs and 650K SVs genotyped in 10K short-read AA samples

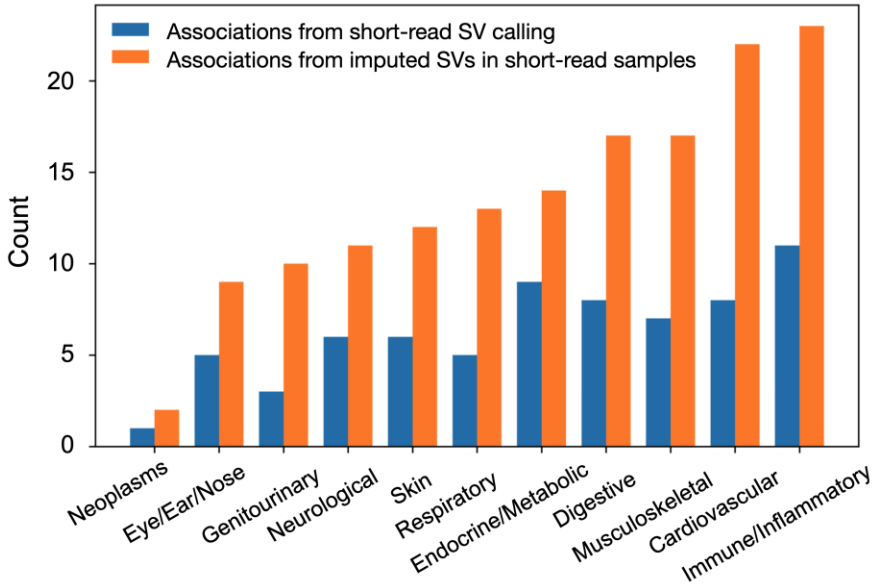
Phenotypes extraction & classification



416 disease phenotypes

Genetic Architecture of Disease Risk

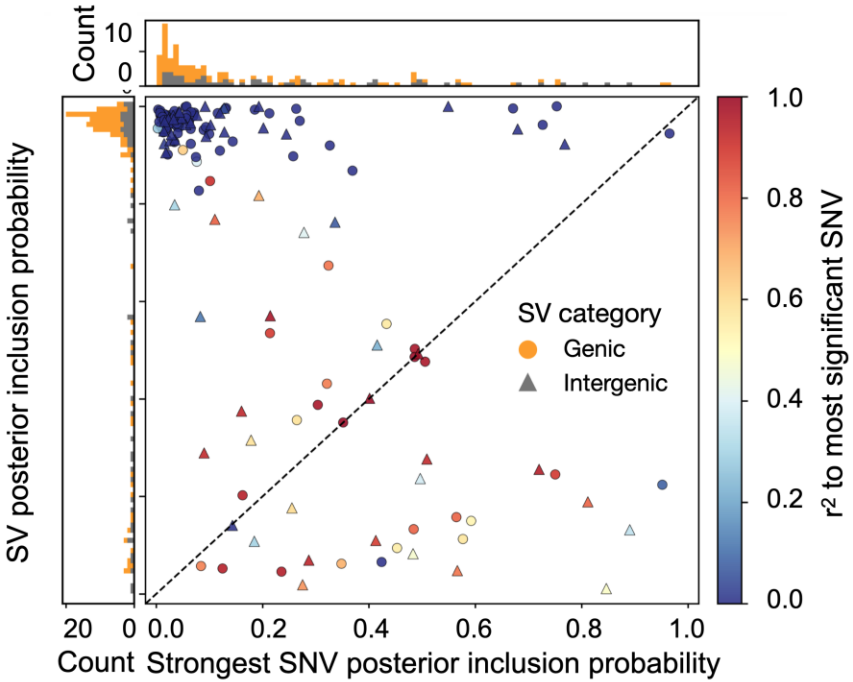




- 150 significant SV-disease associations ($P < 5 \times 10^{-8}$) were identified, spanning 108 distinct conditions
- 81 associations (54%) involve SVs identified only by long-read sequencing and absent from matched short-read callset.

Genome-Wide SV-Disease Associations

Comparison of the association significance between each SV and the strongest nearby SNV (within ±100 kb) for the same phenotype

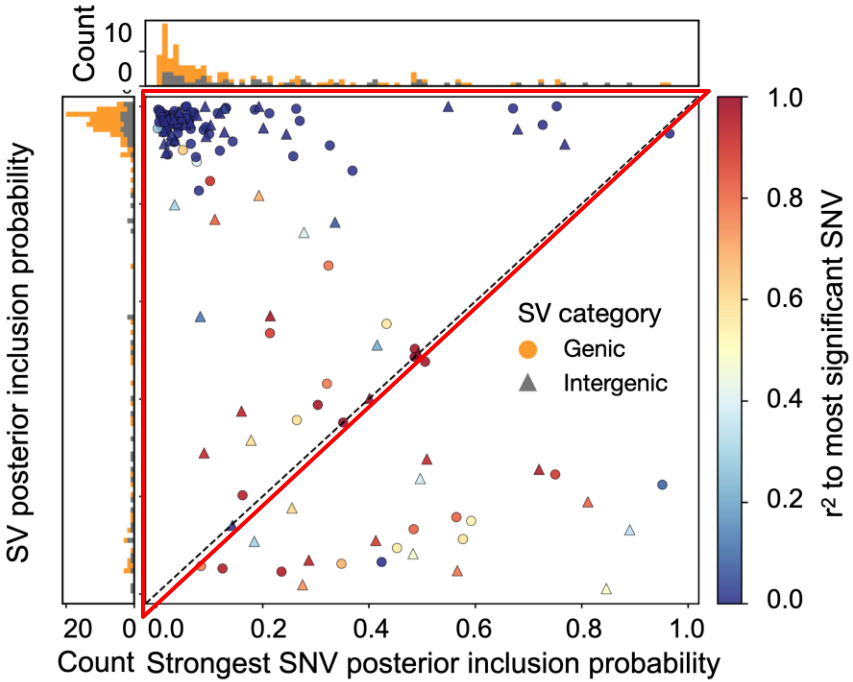


After fine-mapping:

- SVs were included in the 80% casual set in 138 (92.0%) associations.
- 120 associations (80.0 %) where the SV was the most likely causal variant within the locus.

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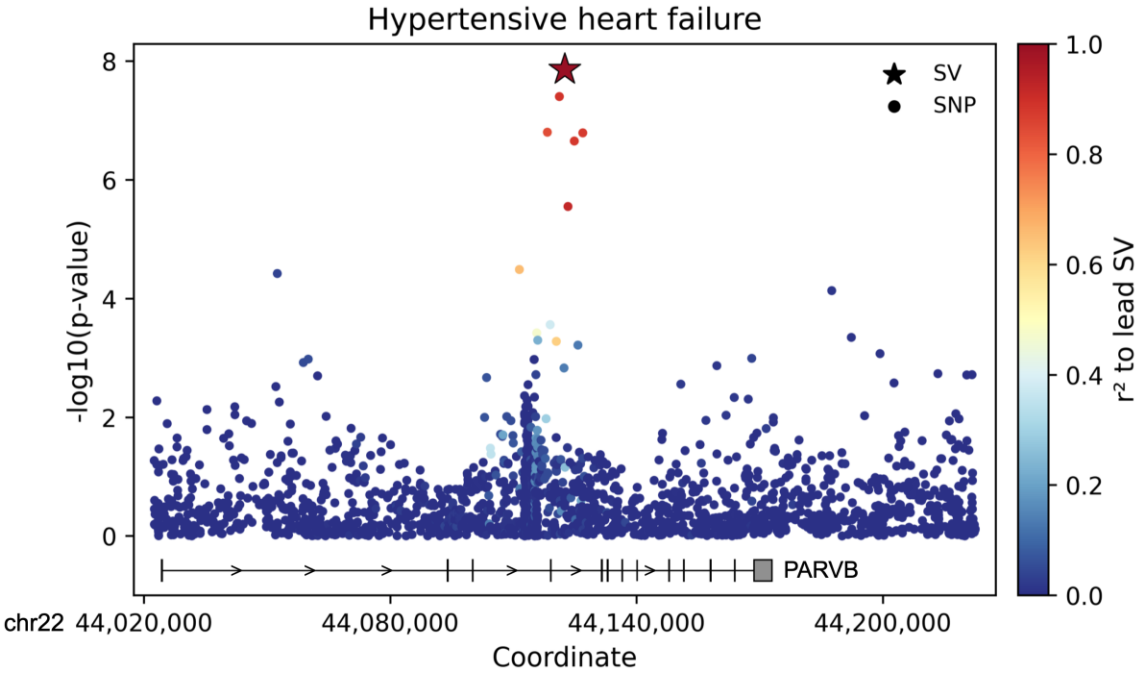
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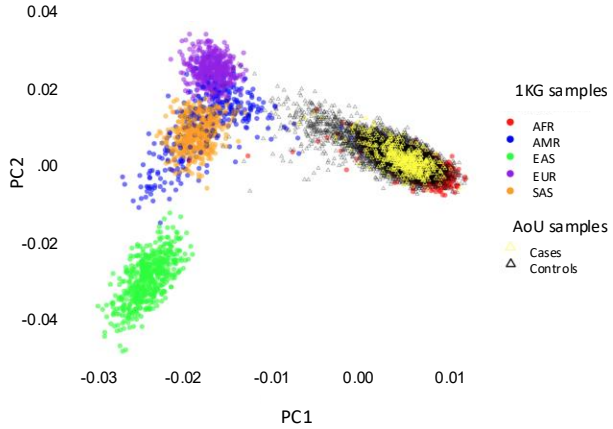
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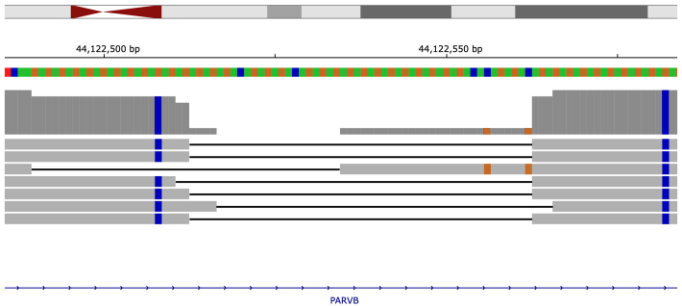
- A 50bp deletion in *PARVB* linked to hypertensive heart failure (OR = 1.76; p = 1.04 × 10^{-8})
- *PARVB* encodes β-parvin, a protein that regulates cardiomyocyte shape and sarcomere assembly
- Independent replication in UK Biobank (431 cases, 153,311 controls): OR = 3.38; p = 0.0033



Samples with hypertensive heart failure: 511



IGV screenshot of the deletion



Thievessen, I. et al. Curr. Biol. 2022

Summary

Comprehensive SV landscape in African Americans

- 68.2% of deletions and 63.4% of insertions missed by short-read WGS

Prioritization of pathogenic variants

- 473 potentially pathogenic variants identified
- 273 absent from previous long-read callsets

SV-eQTL discoveries

- 3,654 significant SV-eQTL associations, including disease- and cancer-related genes
- SVs were the top candidate causal variants for 120 eGenes

GWAS-linked disease signals

- Strong SV-GWAS links across 537 medically relevant traits
- Pleiotropic SV signals across multiple related diseases

Phenome-wide association study

- 150 SV-disease associations identified
  - 81 (54%) absent from matched short-read callsets
  - 120 (80%) were lead variants compared with nearby SNVs

SVs are a major source of disease-relevant variation, contributing to the missing heritability in complex traits and diseases



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Population-scale Long-read Sequencing in the All of Us Research Program

Kiran V Garimella, Qiuhui Li, Julie Wertz, Samuel K Lee, Fabio Cunial, Yongqing Huang, Yulia Mostovoy, Ryan Lorig-Roach, Adam English, Hang Su, Shawn Levy, Donna M Muzny, Chelsea Berngruber, Matt C Danzi, William T Harvey, Emily L LaPlante, Karynne Patterson, Allison N Rozanski, Sophie Schwartz, Beri Shifaw, Yuanyuan Wang, Isaac Wong, Isaac R. L. Xu, Shadi Zaheri, Stephan Zuchner, Xinchang Zheng, Shannon Dugan-Perez, Michal Izydorczyk, Heer Mehta, Richard A Gibbs, Lee Lichtenstein, Namrata Gupta, Niall Lennon, Stacey Gabriel, All of Us Research Program Long Read Working Group, Winston Timp, Kimberly F Doheny, Tara Dutka, Anjene Musick, Chia-Lin Wei, Fritz J Sedlazeck, Michael C Schatz, Michael E Talkowski, Evan E Eichler

doi: https://doi.org/10.1101/2025.10.02.25336942

This article is a preprint and has not been certified by peer review [what does this mean?]. It reports new medical research that has yet to be evaluated and so should *not* be used to guide clinical practice.

Accepted in principle at *Nature*!